MIL-F-50823A (PA)
8 NOVEMBER 1974
SUPERSEDING
MIL-F-50823 (MU)
14 DECEMBER 1972

MILITARY SPECIFICATION

FUZE, GM, ET, LANCE, M811
PARTS, LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, loading, assembling and packing of one type of fuze designated as Fuze, GM, ET, Lance, M811.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-45743 - Soldering, Manual Type, High Reliability Electrical, Electronic Instrument, Communication and Radar for Aerospace and Control Systems, Procedures for

MIL-S-46844 - Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-202 - Test Methods for Electronic and Electric Component Parts

FSC: 1336

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.

MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-810 - Environmental Test Methods
MIL-STD-1168 - Lot Numbering of Ammunition

DRAWINGS

PICATINNY ARSENAL

9251881 - Fuze, M811
9220693 - Drum Metal Shipping for Fuze, GM, Packing and Marking

PUBLICATIONS

NAVORD REPORT 2101 - Statistical Method Appropriate for the Evaluation of Fuze Explosive Train Safety and Reliability

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and Assemblies. - The components and assemblies shall comply with all requirements of the applicable drawings, standards and specifications.

3.3 Requirements for Safety and Arming Device (dwg. 9242183)

3.3.1 Time to Arm. - Both detonators must travel from the safe position to the latched armed position within .80 ± .30 seconds when tested in accordance with 4.4.1.3.

3.3.2 Setback Reset. - The setback weight shall be capable of resetting both rotors from the unlatched armed position to the safe position within 5.0 seconds when tested in accordance with 4.4.1.5.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

3.3.3 Contact Resistance B-D and A-B

3.3.3.1 The resistance between contact pins B and D and contact pins A and B shall not exceed .150 ohms when the rotors are in the armed position when tested in accordance with 4.4.1.4.

3.3.3.2 The resistance between contact pins B and D and contact pins A and B shall not exceed .300 ohms when the rotors are in the safe position when tested in accordance with 4.4.1.1.

3.3.4 Non-Arm Level. - The S&A Device must remain in the safe position when subjected to a steady state acceleration of 8.8 g's + .2 g's for a minimum of 10.0 seconds when tested in accordance with 4.4.1.2.

3.3.5 Insulation Resistance. - The insulation resistance between the following component points shall be greater than 50 megohms when tested in accordance with 4.4.1.6.

K shorted	to	M
L shorted	to	M and K shorted together
F, J and H shorted together	to	Chassis
C and E shorted together	to	Chassis
A, B and D shorted together	to	Chassis
F, J and H shorted together	to	C and E shorted together
F, J and H shorted together	to	A, B and D shorted together
A, B and D shorted together	to	C and E shorted together
P and N shorted together	to	Chassis
K, L, M and R shorted together	to	Chassis

3.3.6 Connector Pin to Pin Resistance. - The resistance, as measured between the specified pins, shall be as follows. Testing shall be in accordance with 4.4.1.7.

Circuit Description	Connector Pins	Resistance
Detonator	A to B	1.5 to 4.5 Ohms
Detonator	B to D	1.5 to 4.5 Ohms
Piston Actuator	F to J	5 to 7.2 Ohms
Piston Actuator	F to H	5 to 7.2 Ohms
Monitor Loop	C to E	0.10 Ohms Max.
Jumper	P to N	0.10 Ohms Max.

3.3.7 Environmental Test

3.3.7.1 Transportation Vibration. - After transportation vibration, the requirements of Paragraph 3.3.6 shall apply. Testing shall be in accordance with 4.4.1.8.1.1.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the currentversion before use.

3.3.7.2 Five Foot Drop. - After five foot drop, the requirements of Paragraph 3.3.6 shall apply. Testing shall be in accordance with 4.4.1.8.1.1.

3.3.7.3 Thermal Shock. - After thermal shock, the requirements of Paragraph 3.3.6 shall apply. Testing shall be in accordance with 4.4.1.8.1.1.

3.3.7.4 Sequence

3.3.7.4.1 Rotor Non-Arm with Lead Out of Line. - After this environment, the requirements of Paragraphs 3.3.4 and 3.3.6 shall apply. Testing shall be in accordance with 4.4.1.8.1.1.

3.3.7.4.2 Rotor Arm with Lead Out of Line. - The requirements of Paragraphs 3.3.1 and 3.3.6 shall apply. Testing shall be in accordance with 4.4.1.8.1.1.

3.3.7.4.3 Missile Vibration. - The S&A Device must remain armed and the requirements of Paragraph 3.3.6 must be satisfied. Testing shall be in accordance with 4.4.1.8.1.1.

3.3.7.4.4 DELETED

3.3.7.4.5 Detonator Functioning. - One or both detonators must function within one (1) millisecond when tested in accordance with 4.4.1.8.1.1.

3.3.8 Jolt. - After being subjected to this environment, no detonator or piston actuator will indicate having functioned when tested in accordance with 4.4.1.8.1.1.

3.3.9 Jumble. - After being subjected to this environment, no detonator or piston actuator will indicate having functioned.

3.3.10 The Safety and Arming Device shall be capable of being stored at temperatures between -65°F and +165°F. The Safety and Arming Device shall operate between -40°F and +140°F.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

3.4 Requirements for Piston Actuator (dwg. 9243602)

3.4.1 No Fire Characteristics

3.4.1.1 Voltage. - No piston actuator shall function when 2.0 + .5 volts DC is applied to the component. Testing shall be in accordance with 4.4.2.7.

3.4.1.2 Current. - No piston actuator shall function under a continuous current of 30 + 5 milliamperes DC applied to the terminals. Testing shall be in accordance with 4.4.2.8.

3.4.2 All Fire Characteristic. - All piston actuators must function when subjected to the minimum all fire voltage (MN-AFV) of 5.0 - .1 VDC in accordance with 4.4.2.6, and at reduced pressure in accordance with 4.4.2.14.

3.4.3 Exhibited Force. - When initiated in accordance with Paragraph 3.4.2, the piston actuator shall be capable of moving a 20 + .5 pound force through a distance of 0.297 inches minimum within 10 milliseconds when tested in accordance with 4.4.2.15.

3.4.4 Identification of Piston Actuator Explosives. - The identity and quantity of all explosive materials used in the component shall be furnished along with the components when submitted for acceptance. No change of material or quantity shall be made without approval of the cognizant Government procuring activity.

3.4.5 Solderability. - The component leadwires shall meet the requirements of MIL-STD-202, Method 208B when tested in accordance with 4.4.2.1.

3.4.6 Terminal Strength. - Lead wires and terminals shall be permanently secured internally and externally, as applicable, to assure that normal movement of the lead wires will not result in damage, wear or excessive strain to the component. The piston actuator, after being tested as specified in 4.4.2.2, shall show no evidence of loose, broken, deformed or displaced parts and shall meet the all fire functional requirements of 4.4.2.6. Force must remain exerted for a minimum of 150 seconds.

3.4.7 Probit Analysis. - When tested and calculated as specified in 4.4.2.5, the all fire voltage shall be less than 5.0 volts DC (VAF) and the no fire voltage shall be greater than 2.0 volts DC (VNF).

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the currentversion before use.

3.4.8 Lead to Lead Resistance. - The DC resistance between the lead wires shall be 6.0 ± 1.0 ohms when tested in accordance with 4.4.2.3.

3.4.9 Insulation Resistance

3.4.9.1 Pre-Fired Condition. - The resistance between the lead wires shunted together and the case shall be 50 megohms minimum at all temperatures. Testing shall be in accordance with 4.4.2.4.

3.4.9.2 Post-Fired Condition. - The resistance between the lead wires shunted together and the case shall be 100 ohms minimum at 140°F. Testing shall be in accordance with 4.4.2.4.

3.4.10 Thermal Shock. - There shall be no evidence of mechanical or electrical damage during or after exposure when tested as specified in 4.4.2.10.

3.4.11 Mechanical Shock. - There shall be no evidence of mechanical or electrical damage during or after exposure when tested as specified in 4.4.2.11.

3.4.12 High Frequency Vibration. - There shall be no evidence of electrical or mechanical damage after exposure to high frequency vibration. Testing shall be in accordance with 4.4.2.12.

3.4.13 Temperature Storage. - There shall be no evidence of electrical or mechanical damage after exposure to -65°F or +160°F with humidity less than 20 percent for indefinite periods.

3.4.14 Modified Temperature and Humidity Cycle (see 4.4.2.13). - The piston actuator shall be capable of meeting all the requirements as stated above after exposure in accordance with Paragraph 4.4.2.13.

3.4.15 Workmanship. - The piston actuator shall be uniform in quality, free from cold soldering, corrosion, pits, cracks, dents, rough edges and defects that will affect life, configuration, serviceability or appearance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

3.5 Requirements for Detonator (9243603)

3.5.1 The detonator shall be capable of being stored at temperatures between -65°F and +160°F for a period of ten years without degradation.

3.5.2 Detonators manufactured more than one year prior to installation into the fuze shall not be used without requalification in accordance with Table I.

3.5.3 Terminal Strength. - The lead wires and terminals shall be permanently secured internally and externally, as applicable, so that normal movement of the lead wires will not result in damage, wear or excessive strain to the component. The detonator shall show no evidence of loose, broken or deplaced parts and shall be capable of meeting the all fire requirements after exposure to the environment called for by 4.4.3.1.

3.5.4 The identity and quantity of all explosive materials used in the detonator shall be furnished with the components submitted for qualification or acceptance testing. No change of material or quantity shall be made without formal approval from the cognizant Government procurement activity.

3.5.5 Solderability. - The detonator lead wires shall meet the requirements of MIL-STD-202, Test Method 208, in the area indicated on dwg. 9243603. Testing shall be in accordance with Paragraph 4.4.3.2.

3.5.6 Functional Requirements

3.5.6.1 Dent Test. - The detonator shall produce an indentation of at least .010 inch deep at the deepest point and the average of the maximum indentation for any group tested shall be at least 0.012 inch. This function must be performed within 400 microseconds. Testing shall be in accordance with Paragraph 4.4.3.6.

3.5.6.2 All Fire Voltage (MN-AFV). - All detonators must function within 400 microseconds maximum with 5.0 VDC applied between terminal points R and R and ground in the circuit shown in Figure 2. Testing shall be in accordance with 4.4.3.5. (NOTE: MN-AFV = Minimum All Fire Voltage.) One detonator may be fired at a time providing a resistor is used to simulate the load of the other detonator.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

3.5.6.3 No Fire Characteristics

3.5.6.3.1 Voltage. - No detonator shall function when 2.5 VDC minimum is applied as V in the test circuit shown in Figure 2

3.5.6.3.2 Current. - No detonator shall function when a current of 50 milliamperes DC minimum (MA-NFC) is continuously applied to its lead wires. Testing shall be in accordance with 4.4.3.8.

NOTE: MA-NFC = Maximum No Fire Current.

3.5.7 Probit Analysis. - This test shall be performed in accordance with 4.4.3.7.

3.5.7.1 Voltage All Fire (VAF). - The all fire voltage shall be equal to or less than 5.0 volts DC. Testing shall be in accordance with 4.4.3.7.1.

NOTE: VAF = Volts All Fire.

3.5.7.2 Voltage No Fire (VNF). - The no fire voltage shall be greater than 2.0 VDC. Testing shall be in accordance with 4.4.3.7.2.

NOTE: VNF = Volts No Fire.

3.5.8 Lead to Lead Resistance. - The resistance between the detonator lead wires prior to initiation shall be 3.5 $\pm$ 1.5 ohms when tested in accordance with 4.4.3.3.

3.5.9 Insulation Resistance. - The resistance between the lead wires shunted together and the case shall be one megohm minimum when tested in accordance with 4.4.3.4.

3.5.10 Thermal Shock. - There shall be no evidence of mechanical or electrical damage during and/or after the exposure to this environment. Testing shall be in accordance with 4.4.3.9.

3.5.11 Modified Temperature and Humidity Cycle. - There shall be no evidence of mechanical or electrical damage during or after the exposure to modified cycle when tested in accordance with 4.4.3.12.

3.5.12 Mechanical Shock. - There shall be no evidence of mechanical or electrical damage during or after exposure to mechanical shock when tested in accordance with 4.4.3.10.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

3.5.13 Vibration. - There shall be no evidence of mechanical or electrical damage during or after exposure to the vibration environment when tested as specified in 4.4.3.11.

3.5.14 Moisture Resistance. - There shall be no evidence of mechanical or electrical damage during or after exposure to moisture environment.

3.5.15 Reduced Pressure. - The detonator shall meet the all fire voltage requirement at reduced pressure. Testing shall be in accordance with 4.4.3.13.

3.5.16 Workmanship. - The detonator shall be manufactured and processed in such a manner as to be uniform in quality, free from cold soldering, corrosion, pits, cracks, dents, rough edges and other defects that could affect life, configuration serviceability or appearance.

3.6 Requirements for Fuze Pre-Assembly and Assembly Checkout (see 4.4.4)

3.6.1 The safety and Arming (S&A) Device shall show no evidence of foreign materials.

3.6.2 Setback Reset. - The "g" weight shall be free to move when tested in accordance with 4.4.4.1.

3.6.3 All soldering shall conform to MIL-S-45743 and MIL-S-46844.

3.6.4 P Pin to Pin Resistance. - The Safety and Arming Device shall exhibit the following pin to pin resistances between receptacle contacts:

Circuit Description	Connector Pins	Requirement
Piston Actuator	F to J	5.7 to 7.2 Ohms
Piston Actuator	F to H	5.7 to 7.2 Ohms
Detonator	B to A	1.5 to 4.5 Ohms
Detonator	B to D	1.5 to 4.5 Ohms
Breakwire	C to E	.10 Ohms Max.

Testing shall be in accordance with 4.4.4.3.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.

3.6.5 Housing Assembly

3.6.5.1 SECO Arm Event Test. - SECO, Arm and Event times shall be within .07% of the settings. For the XM811E6 time representation of the SECO letter settings are as follows:

XM811E6	Letter
55 Seconds	A
65 Seconds	B
72 Seconds	C
77 Seconds	D
83 Seconds	E
90 Seconds	F
100 Seconds	G
110 Seconds	H
120 Seconds	J
Counter Remains On Indefinitely	OFF

(Terminate test after Event occurs)

Testing shall be in accordance with 4.4.4.4.

3.6.5.2 Insulation Resistance. - The insulation resistance of the following pins to the housing shall be 10 megohms minimum when measured at VDC with the current limited to one milliamp:

All Pins on P2 and J3 Connectors

3.6.5.3 Arm, SECO and Event Firing Circuits. - The discharge characteristic of each of the Arm, SECO and Event firing circuits Pins J3-H, J3-J, J3-A, J3-D, P2-6 and P2-14, shall be within the limits shown in Figure 5 when measured across a 100 ohm resistor placed between each of the above pins respectively and P2-13. The capacitors shall be charged to 5 VDC for the test which can be done by applying VDC continuously between pins P2-15 (positive) and P2-13 (ground) and allowing a minimum of 10 seconds prior to taking readings. At the conclusion of the test, the voltage shall be removed from pin P2-15 and pin P2-15 shall be shorted to P2-13 to insure that there is no charge on the capacitors - for 10 seconds minimum.

3.6.5.4 Pin to Pin Resistance. - The housing subassembly shall meet the requirements of the following pin to pin resistance values. This test shall be conducted using five (5) volts or less with current limited to one milliamp.*

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the currentversion before use.

Circuit Description	Connector Pins	Resistance
Set Interconnect	P2-1 to P2-7	1K ± 10%
Event Charging Circuit	P2-15 (Pos) to J3-A	15K ± 25%
Event Charging Circuit	P2-15 (Pos) to J3-D	15K ± 25%
Arming Charging Circuit	P2-15 (Pos) to J3-H	15K ± 25%
Arming Charging Circuit	P2-15 (Pos) to J3-J	15K ± 25% **
SCO Charging Circuit	P2-15 (Pos) to P2-6	32K ± 25%
SCO Charging Circuit	P2-15 (Pos) to P2-14	32K ± 25%
Event Diode	P2-15 to J3-A (Pos)	1 Megohm Min.
Event Diode	P2-15 to J3-D (Pos)	1 Megohm Min.
Arm Diode	P1-15 to J3-H (Pos)	1 Megohm Min.
Arm Diode	P2-15 to J3-J (Pos)	1 Megohm Min.
SCO Diode	P2-15 to P2-6 (Pos)	1 Megohm Min.
SCO Diode	P2-15 to P2-14 (Pos)	1 Megohm Min.
B+ Interconnect	P2-7 to J3-K	0.1 Ohm Max.
Gnd Interconnect	P2-13 to J3-L	0.1 Ohm Max.
SCO Interconnect	P2-5 to J2-N	0.1 Ohm Max.
Set Interconnect	P2-1 to J3-M	0.1 Ohm Max.
Event T/M	J3-B to J3-C (Pos)	12.5K ± 25%
Event T/M	J3-B (Pos) to J3-C	1 Megohm Min.
Arm T/M	J3-F to J3-C (Pos)	15K ± 25%
Arm T/M	J3-F (Pos) to J3-C	1 Megohm Min.

* AC measurements shall be one (1) ma RMS max.
** Short P2-15 to P2-13 for 10 seconds minimum after this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

3.6.6 Fuze Assembly Checkout

3.6.6.1 Insulation Resistance. - The insulation resistance between the following points shall be 50 megohms minimum when tested at 50 VDC with the current limited to one (1) milliamp.

J1-1	to	Chassis
J1-5	to	Chassis
J1-6	to	J1-2
J1-6	to	Chassis
J1-2	to	Chassis
J1-7	to	Chassis
J1-11	to	Chassis

3.6.6.2 Pin to Pin Resistance. - The fuze shall meet the requirements of the following pin to pin resistance values. The test shall be conducted using five (5) volts or less and limiting the current to one (1) milliamp. *

Circuit Description	Connector Pins	Resistance
Battery Match	J1-1 to J1-8	2.4 to 2.7 Ohms
Continuity Loops	J1-11 (Pos) to J1-14	3K ± 40%** Ohms
SCO Squib Switch	J1-4 to J1-11 (Pos)	5K ± 40%** Ohms
SCO Output	J1-6 to J1-10	50 Megohms Min.
SCO Interlock	J1-5 to J1-6	50 Megohms Min.
Jumper	J1-1 to J1-9	.1 Ohms Max.
Jumper	J1-8 to J1-15	.1 Ohms Max.
Jumper	J1-6 to J1-13	.1 Ohms Max.
Jumper	J1-2 to J1-10	.1 Ohms Max.
Jumper	J1-3 to J1-4	.1 Ohms Max.
Jumper	J1-14 to J1-7	.1 Ohms Max.
Jumper	J1-14 to J1-12	.1 Ohms Max.
* SCO Interlock	J1-5 to J1-10	50 Megohms Min.

** AC measurements shall be one (1) ma RMS max.
*** 10 ma max. this test only.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

3.7 Requirements for Battery, Reserve

3.7.1 Case Change. - During and after the discharge life of the component, the terminals shall not extend more than 0.24 inches from the edge of the case when tested in accordance with 4.4.5.18.

3.7.2 Terminal Strength. - All terminal and lead wire elements shall be secured internally and externally as applicable, so that normal movement of these items will not result in electrical damage, mechanical damage, wear or excessive strain to the elements, enclosure or coating when tested in accordance with 4.4.5.1.

3.7.3 Seal. - The leak rate shall not exceed 1 x 10 ATM cc/sec of helium at 25°C when tested in accordance with 4.4.5.2.

3.7.4 Deleted

3.7.5 Initiation Characteristics

3.7.5.1 All Fire. - The battery shall initiate and activate when a 3.5 ampere pulse of 25 milliseconds maximum is applied to the match lead wires when tested in accordance with 4.4.5.4.

3.7.5.2 No Fire. - The battery shall not initiate or function when a 1.0 ampere pulse of three (3) seconds minimum is applied to the match lead wires when tested in accordance with 4.4.5.5.

3.7.6 Match (Bridgewire) Resistance. - The pre-initiation resistance shall be 1.0 ± .25 ohms when tested in accordance with 4.4.5.6.

3.7.7 Insulation Resistance. - The resistances between the output terminals and the bridgewire leads, between the output terminals and the case, and between the bridgewire leads and the case shall be greater than 10 megohms when tested in accordance with 4.4.5.7.

3.7.8 Pre-Initiation Voltage. - The output voltage measured between the output terminals shall be 0.15 VDC maximum when tested in accordance with 4.4.5.8.

3.7.9 Polarity. - The polarity of the voltage shall conform to the markings on the battery case when tested in accordance with 4.4.5.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

3.7.10 Output Voltage. - The voltage of the battery, when connected to a fixed resistive load and tested as specified in 4.4.5.10, shall rise to 6.1 VDC within 2.35 seconds after initial pulsing of the match and shall remain between 6.5 and 10.0 VDC for 210 seconds minimum. The voltage shall not exceed 10.0 VDC at any time.

3.7.11 Noise Voltage. - Noise appearing as a spurious voltage in the output of the battery, when connected to a fixed resistive load and tested as specicied in 4.4.5.11, shall not exceed 250 millivolts peak to peak within a frequency band of 80 to 2000 Hertz.

3.7.12 Thermal Shock. - The battery shall withstand exposure to thermal shown between the temperatures of -40°F and +140°F. Testing shall be in accordance with 4.4.5.12.

3.7.13 Humidity. - The battery shall withstand exposure to a 95% relative humidity. Testing shall be in accordance with 4.4.5.13.

3.7.14 The battery shall withstand exposure to the vibration test in accordance with MIL-STD-810B with the following exceptions. The units shall be subjected to vibration over the frequency range of five (5) to 500 HZ at a double amplitude of 0.5 inch or 3.5g's peak, whichever is less. Sweep time from 5 to 500 to 5 HZ shall be 15 minutes and shall be a logrithmic sweep. The total duration of vibration shall be four (4) hours in each of the three mutually perpendicular axis. The units to be tested cold shall be conditioned at -65°F and the units to be tested hot shall be conditioned at +160°F.

3.7.15 Flight Vibration. - The battery shall withstand vibration between five (5) and 2000 Hertz at 10g's acceleration. Testing shall be in accordance with 4.4.5.15.

3.7.16 Acceleration. - The battery shall withstand exposure to a sustained acceleration of 4g's along the longitudinal axes and an acceleration of 40g's normal to the longitudinal axis. Testing shall be in accordance with 4.4.5.16.

3.7.17 Mechanical Shock. - The battery shall withstand exposure to 50g, 10 millisecond shock pulses when tested in accordance with 4.4.5.17.

3.7.18 Temperature Storage. - The battery shall be capable of withstanding storage at temperatures ranging between -65°F and +160°F without evidence of damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

3.7.19 Workmanship. - The battery shall be manufactured and processed in such a manner as to be uniform in quality, free from cold soldering, corrosion, pits, cracks, dents, rough edges and other defects that might affect life, configuration, serviceability or appearance of the item.

3.7.20 Battery Pre-Initiation Internal Resistance. - The internal resistance of the battery, when measured between the positive and negative battery terminals, shall be 100K ohms minimum. Test voltage shall be 50 VDC maximum. Test current shall be one (1) milliampere maximum.

3.8 Requirements for Switch, Explosive, Multi-Circuit (see dwg. 9243595)

3.8.1 Terminal Strength. - The lead wires and terminals shall be permanently secured internally and externally, as applicable, to assure that normal movement of the lead wires will not result in damage (electrical or mechanical), wear or excessive strain to the component. The explosive switch, after being tested as specified in 4.4.6.1, shall show no evidence of loose, broken, deformed or displaced parts and shall meet the all fire requirement specified in 3.8.3.

3.8.2 Solderability. - The component lead wires shall meet the requirements of MIL-STD-202, Method 208 when tested in accordance with 4.4.6.2.

3.8.3 All-Fire Characteristics. - The switch shall be caused to initiate by the discharge of a capacitor in accordance with 4.4.6.3. After functioning, the switch configuration shall meet the post-fired requirement of Figure 1 within three (3) milliseconds on the first try and fail to meet all the pre-fired requirements.

3.8.4 No-Fire Characteristics. - The switch shall not function when exposed to a constant DC current of 40 milliamperes through the unshunted lead wire. After exposure, the switch shall meet all the pre-fired requirements of Figure 1 and shall fail to meet all the post-fired requirements. Testing shall be in accordance with 4.4.6.4.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

3.8.5 Input (Bridgewire) Resistance. - The DC resistance between the component input leads A and B prior to initiation shall be 12 ± 2.0 ohms when tested as specified in 4.4.6.5.

3.8.6 Output (Switch) Resistance. - The pre-fired resistance between the switch terminals (of the individual switch sections) shall be:

1. 100 Megohms minimum at 500 volts for the open circuit switch contacts (see Figure 1).

2. 0.008 Ohms maximum at one (1) amp for the closed circuit switch contacts (see Figure 1).

The post-fired DC resistance between the switch terminals (of the individual switch sections) shall be:

1. 100 Megohms minimum at 500 volts for the open circuit switch contacts (see Figure 1).

2. 0.008 Ohms maximum at one (1) amp for the closed circuit switch contacts (see Figure 1).

All testing shall be in accordance with 4.4.6.6.

3.8.7 Insulation Resistance. - The resistance shall be 100 megohms minimum prior to initiation and shall be 20 megohms minimum after initiation. Testing shall be in accordance with 4.4.6.7.

3.8.8 Resistance. - The switch circuits shall exhibit the appropriate state, i.e., open or closed, as required by Figure 1. A closed circuit shall exhibit a resistance of .1 ohm maximum; an open circuit shall exhibit a resistance of 50 megohms minimum. Testing shall be in accordance with 4.4.6.8.

3.8.9 Thermal Shock. - There shall be no evidence of mechanical or electrical damage during or after exposure when tested as specified in 4.4.6.9.

3.8.10 Mechanical Shock. - There shall be no evidence of mechanical or electrical damage during or after exposure. The output circuit contact chatter shall not exceed 1.0 microseconds. Testing shall be in accordance with 4.4.6.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

3.8.11 Vibration. - The switch shall withstand exposure to sinusoidal vibration without evidence of mechanical or electrical damage. The output circuit contact chatter shall not exceed 1.0 microseconds. Testing shall be in accordance with 4.4.6.11.

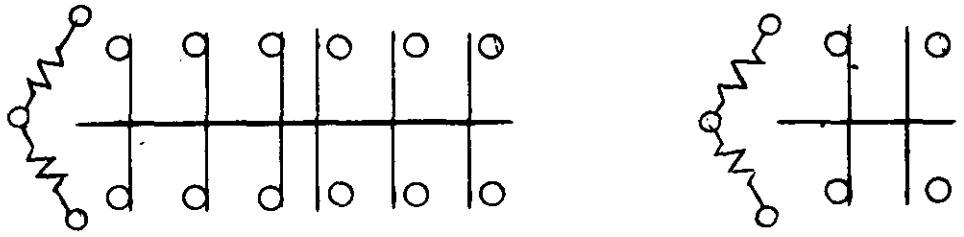
3.8.12 Temperature Storage. - The switch shall function satisfactorily as evidenced by meeting the requirements of 3.8.3 after exposure to the environments as specified in 4.4.6.12. The component shall show no evidence of damage.

3.8.13 Temperature and Humidity Cycle. - The switch shall function satisfactorily as evidenced by meeting requirements of 3.8.3 after exposure to the environments as specified in 4.4.6.13.

3.8.14 Workmanship. - The component shall be manufactured and processed in such a manner as to be uniform in quality, free from cold soldering, corrosion, pits, cracks, dents, rough edges and other defects that could affect life, configuration, serviceability or appearance.

FIGURE 1

SWITCH, EXPLOSIVE, MULTI-CIRCUIT
CONTACT ARRANGEMENT PRE-FIRE CONDITION (SEE DWG. 9243595)



Part No.	No. of Circuits	Pre-Fire		Post-Fire	
		Open	Closed	Open	Closed
1	6	S1, S2 & S3	S4, S5 & S6	S4, S5 & S6	S1, S2 & S3
2	2	S1	S2	S2	S1

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

3.8.15 Cover Assembly Checkout

3.8.15.1 Insulation Resistance. - The insulation resistance from the following pins to chassis ground shall be 50 megohms minimum when measured at 50 VDC with the current limited to one (1) milliamp.

J2-1, J2-3, J2-5, J2-2, J2-15, J2-9 and J2-10

3.8.15.2 Firing Circuits - BCO - With five (5) VDC applied to pins J2-7 (positive) and J2-13 (negative), the charging current, as measured across a 100 ohm resistor in series with J2-7 and the five (5) VDC supply, shall be within the limits shown in Figure 6.

CAUTION: J2-4 must be left floating during this test to avoid initiating the squib switch.

NOTE: At the conclusion of the above tests, J2-7 should be shorted to J2-13 for a minimum of 30 seconds to discharge the capacitors.

3.8.15.3 Pin to Pin Resistance. - The assembly shall meet the requirement of the following pin resistive values. This test shall be conducted using five (5) volts or less and limiting the current to one (1) milliamp or less. *

Circuit Description	Connector Pins	Resistance
Battery Matches	J2-2 to J2-12	.4 to .7 Ohms **
SCO Bridgewire	J2-5 to J2-6	5 to 7.5 Ohms
SCO Bridgewire	J2-5 to J2-14	5 to 7.5 Ohms
Resistor	J2-15 to J2-13	10K Ohms ± 20%
SCO	J2-9 to J2-10	50 Megohms Min.
SCO Interlock	J2-9 to J2-3	50 Megohms Min.
Set Contacts	J2-1 to J2-13	50 Megohms Min.
Monitor Loop	J2-8 to J2-13	12K ± 10%
Monitor Loops	J2-8 to J2-7	2K ± 10% **
BGG Squib Circuit	J2-4 to J2-7 (Pos)	8 K ± 25%
BGG Diode Check	J2-4 (Pos) to J2-7	1 Meghom Min.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

* AC measurements shall be one (1) ma RMS max.
** Ten (10) ma max. this test only.

NOTE: At conclusion of above tests, J2-7 and J2-15 shall be shorted to J2-13 for 30 seconds minimum.

3.8.15.4 Cover Assembly Pre-Potting Checkout. - The following measurements shall be made prior to potting the cover assembly:

Battery Diode	Terminal 22 (Pos) to J2-7	500 Ohms ± 30% *
Battery Diode	Terminal 23 (Pos) to J2-7	500 Ohms ± 30% *
**Battery Diode	Terminal 22 to J2-7 (Pos)	6 Kilohms Min.
**Battery Diode	Terminal 23 to J2-7 (Pos)	6 Kilohms Min.

* Ten (10) ma max.
** Battery terminals to be shorted during this test.

3.9 Fuze Environmental Checkout. - The fuze shall exhibit the proper functioning when tested in accordance with Paragraph 4.4.7. Proper functioning shall consist of:

a. Time outs within .07% of set time for SECO, Arm and Event.
b. Proper functioning of explosive train components.
c. S&A rotors fully arming under a 14g $^{+0}_{-1}$ load within 10 seconds.

3.10 Rotary Switch (9208789-1 and 9208789-2). - Contacts shall measure ten (10) milliohms maximum with a maximum measuring current of 100 ma.

3.11 Workmanship - Parts and Assemblies. - The parts and assemblies shall be free from chips, dirt, grease and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

3.12 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the currentversion before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of Product. - At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with an identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample to a Government approved facility as designated by the Contracting Officer (see 6.1) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

Part No.	Description	Quantity
9243615-3	Standoff	10
9248637	Rotor Lead Cup Assembly	10
9242217	Rotor Lead Cup	10
9208811	Lead Cup Assembly	10
9243610	Terminal Board Assembly	10
9243618	Contact	10
9243611	Contact	10
9251886	Terminal Board	10
9242216	Base	25
9251806	Connector Receptacle	10
9243603	Detonator	490
9243595	Switch Explosive Multi-Circuit	115
9242199	Cover Sub-Assembly	5
9243628	Standoff Connector	10
9242182	Cover Assembly	5
9242188	Housing	30
9251881	Fuze, XM811E6, Less S&A	12
9242183	Safety and Arming Devices	28
9243608	Battery	14
9248625	Standoff Connector	10
9242195	Printed Wiring Board Assembly (A1)	5
9251883	Printed Wiring Board Assembly (A2)	5
9251884	Printed Wiring Board Assembly (A3)	5
9242193	Printed Wiring Board Assembly (A4)	5

All items included in the first article sample shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and shall submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials (see 4.1.1.c.). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:03Z
Check the source to verify that this is the currentversion before use.

4.2.2 Inspections to be Performed. - Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and to any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any component or subassembly fails to comply with the applicable requirement, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component, subassembly or assembly in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection Provisions

4.3.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Standoff (see dwg. 9243615-3 covering a detail of dwg. 9242183).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.10 Percent	
	101. Overall length ('A') dimension	Gage
Minor:	AQL 0.65 Percent	
	201. Evidence of poor workmanship	Visual

4.3.2.2 Rotor Lead Cup Assembly (see dwg. 9248637 covering a detail of dwg. 9242183).

Categories	Defects	Method of Inspection
Critical:		
	1. Shear pin is not present or staked	Visual
	2. True position of cross pin	Gage
Major:	None defined.	
Minor:	AQL 0.65 Percent	
	201. Evidence of poor workmanship	Visual

4.3.2.3 Rotor Lead Cup (see dwg. 9242217 covering a detail of dwg. 9248637).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.40 Percent	
	101. Circumferential profile	Comparator
	102. Position of lead receptacles	Gage
	103. Location of cross pin hole	Gage

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the currentversion before use.

104.	Large diameter of cup seating ledge	Gage
105.	Small diameter of cup seating ledge	Gage

Minor: AQL 0.65 Percent

201.	Evidence of poor workmanship	Visual

4.3.2.4 Terminal Board Assembly (see dwg. 9243610 covering a detail of dwg. 9242183).

Categories	Defects	Method of Inspection

Critical: None defined.

Major: AQL 0.25 Percent

101.	Contacts secured incorrectly	Manual
102.	Contacts at improper angle	Gage
103.	Contacts incorrectly positioned or oriented	Gage/ Visual

Minor: AQL 0.65 Percent

201.	Evidence of poor workmanship	Visual

4.3.2.5 Contact (see dwg. 9243618 covering a detail of dwg. 9242183).

Categories	Defects	Method of Inspection

Critical: None defined.

Major: AQL 0.25 Percent

101.	Lack of or improper gold plating, min.	Microscope
102.	Position of eyelet hole	Gage
103.	Depth of recess	Gage
104.	Length of recess	Gage
*105.	Thickness of gold plate	Gage

Minor: AQL 0.65 Percent

201.	Evidence of poor workmanship	Visual

* A random selection of five (5) percent of these units shall be subjected to analysis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.2.6 Contact (see dwg. 9243611 covering a detail of dwg. 9212183).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.15 Percent	
101.	Position of eyelet hole	Gage
102.	Lack of or improper gold plating ..	Microscope
*103.	Thickness of gold plating	Gage
Minor:	AQL 0.65 Percent	
201.	Evidence of poor workmanship	Visual

* A random selection of five (5) percent of these units shall be subjected to analysis.

4.3.2.7 Terminal Board (see dwg. 9251886 covering a detail of dwg. 9242183).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.40 Percent	
101.	Position of eyelet holes	Gage
102.	Width of contact channels	Gage
103.	Height overall (1.34 - .01)	Gage
104.	Relief for standoffs, depth of	Gage
105.	Rounded corners and shoulders on wire channel	Visual
106.	Set screw hole thread minor diameter	Gage
Minor:	AQL 0.65 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.8 Base (see dwg. 9242216 covering a detail of dwg. 9232183).

Categories	Defects	Method of Inspection
Critical:		
1.	Thickness of lead blow through parts, min. (2 places)	Gage

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.65 Percent	
101.	Height of shear wire seating post	Gage
102.	Depth of shear wire slot	Gage
103.	Height of base	Gage
104.	Thickness of flange	Gage
105.	Diameter of rotor lead cup receptacle	Gage
106.	Profile of blow through ports	Gage
107.	Depth of connector standoff screw holes	Gage
108.	Depth of terminal board standoff screw holes	Gage
109.	Surface finish of rotor, lead cup receptacle side wall	Gage
Minor:	AQL 0.65 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.9 Connector Receptacle (see dwg. 9251086 covering a detail of dwg. 9242183).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.25 Percent	
101.	Socket contacts identified incorrectly	Visual
102.	Socket contact improper size	Gage
103.	Guides incorrectly assembled	Visual
Minor:	AQL 0.65 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.10 Deleted

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.2.12 Switch, Explosive, Multi-Circuit (see dwg. 9243595 covering a detail of dwg. 9242199).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.25 Percent	
101.	Terminals numbered improperly	Visual
102.	Width improper	Gage
103.	Width of terminals improper	Gage
104.	Location of terminals improper	Gage
Minor:	AQL 0.65 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.13 Cover Sub-Assembly (see dwg. 9242199 covering a detail of dwg. 9242182).

Categories	Defects	Method of Inspection
Critical:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

Major: 100 Percent Inspection
101. PWB-A1 or J2 connector terminals connected to incorrect plate ..... Visual
102. Color of wires incorrect ..... Visual
103. Printed wiring board (PBW) A-1 not secured properly ..... Manual
104. Soldering not in accordance with MIL-S-60902 ..... Visual

Minor: AQL 0.65 Percent
201. Evidence of poor workmanship ..... Visual

4.3.2.14 Standoff, Connector (see dwg. 9243628 covering a detail of dwg. 9242199).

Categories Defects Method of Inspection

Critical: None defined.

Major: AQL 0.15 Percent
101. Overall length ..... Gage
102. Screw thread lengths (2 places) ... Gage

Minor: AQL 0.65 Percent
201. Evidence of poor workmanship ..... Visual

4.3.2.15 Cover Assembly (see dwg. 9242182 covering a detail of dwg. 9251881).

Categories Defects Method of Inspection

Critical: None defined.

Major: 100 Percent Inspection
101. Wiring damaged or disconnected .... Visual
102. Potting compound on J2 connector terminals ..... Visual

Major: AQL 0.15 Percent
103. Potting compound in unauthorized area ..... Visual
104. Connector orientation improper .... Visual

Minor: AQL 0.65 Percent
201. Evidence of poor workmanship ..... Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.2.16 Housing (see dwg. 9242188 covering a detail of dwg. 9251880).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.40 Percent	
101.	Position of missile mating holes (4 places)	Gage
102.	Diameter of missile mating holes	Gage
103.	Cover mating hole thread minor diameter, max.	Gage
104.	Pitch diameter of cover mating holes	Gage
Minor:	AQL 1.0 Percent	
201.	Minor diameter of cover mating holes, min.	Gage
202.	Evidence of poor workmanship	Visual

4.3.2.17 Printed Wiring Board Assembly (A1) - (see dwg. 9242195 covering a detail of dwg. 9242199).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	100 Percent Inspection	
101.	Soldering not in accordance with MIL-S-60902	Visual
102.	Cracks present	Visual
Minor:	AQL 0.40 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.18 Printed Wiring Board Assembly (A2) - (see dwg. 9251883 covering a detail of dwg. 9251879).

Categories	Defects	Method of Inspection
Critical:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

Major:	100 Percent Inspection	
101.	Soldering not in accordance with MIL-S-60902	Visual
102.	Cracks present	Visual
Minor:	AQL 0.40 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.19 Printed Wiring Board Assembly (A3) - (see dwg. 9242193 covering a detail of dwg. 9251879).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	100 Percent Inspection	
101.	Soldering not in accordance with MIL-S-60902	Visual
102.	Cracks present	Visual
Minor:	AQL 0.40 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.20 Printed Wiring Board Assembly (A4) - (see dwg. 9241883 covering a detail of dwg. 9251879).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	100 Percent Inspection	
101.	Soldering not in accordance with MIL-S-60902	Visual
102.	Cracks present	Visual
Minor:	AQL 0.40 Percent	
201.	Evidence of poor workmanship	Visual

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.2.21 Safe and Arm Assembly (A5) - (see dwg. 9242183 covering a detail of dwg. 9251881).

Categories	Defects	Method of Inspection
Critical:		
1.	Rotor in line (armed)	Visual
2.	Weight, setback missing	Visual
3.	Safe and arm indicator missing, improperly secured, or extending below the base of the G weight	Visual
Major:	AQL 0.65 Percent	
101.	Contact force	Gage
102.	Connector orientation	Visual
103.	Wire potting not flush or below top of terminal board	Visual
104.	Lack of shrinkable tubing on actuator leads	Visual
105.	Lack of insulation tubing on all connector leads and terminals	Visual
106.	Proper lead wires connected to improper connector terminals	Visual
107.	Screws not below top terminal board	Visual
108.	Minimum torque on screws	Gage
109.	Connector not parallel to base (not called for)	Visual
110.	Excess lubricant	Visual
Minor:	AQL 0.40 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.22 Standoff, Connector (see dwg. 9248625 covering a detail of dwg. 9242183).

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:	AQL 1.5 Percent	
101.	Overall length	Gage
102.	Hexagonal length	Gage
103.	Depth of threaded receptacle	Gage
Minor:	AQL 0.40 Percent	
201.	Evidence of poor workmanship	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.2.23 Lead Cup Assembly (see dwg. 9208811 covering a detail of dwg. 9236837).

Categories	Defects	Method of Inspection
Critical:		
1.	Explosive beyond surface A	Visual
Major:	AQL 0.25 Percent	
101.	Maximum height	Gage
102.	Large diameter, max.	Gage
103.	Minimum seating flange (not called for)	Gage
Minor:	AQL 0.65 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.24 Fuze, Guided Missile, XM811E6 (see dwg. 9251881).

Categories	Defects	Method of Inspection
Critical:		
1.	Safety and Arming Device not in the safe position	Visual
Major:	100 Percent	
101.	Marking incorrect	Visual
102.	Identification plate incorrect	Visual
103.	Part missing	Visual
Major:	AQL 0.25 Percent	
104.	Screw torque improper, min.	Gage
Minor:	AQL 0.40 Percent	
201.	Evidence of poor workmanship	Visual

4.3.2.25 Fuze, Guided Missile, XM811E6, Packaging (see dwg. 9220693).

Categories	Defects	Method of Inspection
Critical: None defined.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

Major:	100 Percent	
101.	Improperly packed	Visual
Minor:	AQL 0.40 Percent	
201.	Evidence of poor workmanship	Visual

4.3.3 Testing. - All test and inspection equipment used for the performance of all the following tests shall be in accordance with Paragraph 4.3.4.

4.3.3.1 First Article Sample (see 4.2.1). - The components and assemblies submitted as the first article sample shall be tested as specified below after having been examined for the requirements as specified in 4.3.2. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.3.1.1 Safety and Arming Device

4.3.3.1.1.1 Contact Resistance B-D and A-B (see 3.3.3.2) - Major Defect. - This test shall be performed 100 percent. Any Safety and Arming Device failing to meet the requirement shall be cause for rejection of the first article sample. Testing shall be in accordance with 4.4.1.1.

4.3.3.1.1.2 Connector Pin to Pin Resistance (see 3.3.6) - Major Defect. - Twenty-four (24) Safety and Arming Devices shall be subjected to this test as specified in Paragraph 4.4.1.8. Any assembly failing to meet the requirements of 3.3.6 shall be cause for rejection of the first article sample.

4.3.3.1.1.3 Non-Arm Level (see 3.3.4) - Major Defect. - Twenty-four (24) Safety and Arming Devices shall be subjected to this test as specified in Paragraph 4.4.1.2. Any assembly failing to meet the requirements shall be cause for rejection of the first article sample.

4.3.3.1.1.4 Time to Arm (see 3.3.1) - Major Defect. - Twenty-four (24) Safety and Arming Devices shall be subjected to this test as specified in Paragraph 4.4.1.3. Any assembly failing to meet the requirement shall be cause for rejection of the first article sample.

4.3.3.1.1.5 Contact Resistance (see 3.3.3.1) - Major Defect. - Twenty-four (24) Safety and Arming Devices shall be subjected to this test as specified in Paragraph 4.4.1.4. Any assembly failing to meet the requirement shall be cause for rejection of the first article sample.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.1.1.6 Setback Reset (see 3.3.2) - Major Defect. - This test shall be performed on 24 Safety and Arming Devices. Failure of any unit to meet the requirement shall be cause for rejection of the first article sample. Testing shall be in accordance with 4.4.1.5.

4.3.3.1.1.7 DELETED

4.3.3.1.1.8 Jolt (see 3.3.8) - Major Defect. - Assemblies for this test shall consist of two (2) Safety and Arming Devices mated to two (2) housings with appropriate mating connector and leads potted in place and appropriate ballast secured to simulate the end item fuze weight. Any fuze not meeting the requirements shall be cause for rejection of the first article sample. Testing shall be in accordance with 4.4.1.8.1.1.

4.3.3.1.2 Detonator

4.3.3.1.2.1 First Article and Qualification Test. - The sample sizes for this test shall be as specified in the following table. The detonators shall be subjected to the examinations and tests in the order shown in the table. The total sample detonators shall be selected and divided into groups as specified and subjected to the inspections for their respective group. Test procedures will be as specified in the applicable paragraph as shown. Detonators used in this test will not be submitted as a delivered quantity against the contract for intended use in final assembled fuzes. Items shall be tested at the temperatures indicated unless otherwise specified in the procedural paragraph or referenced document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 1

DETONATOR FIRST ARTICLE AND QUALIFICATION TESTS

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	1/ °F + 5°F +75	1/ °F + 5°F +140	2/	3/
GROUP I						0	30
WORKMANSHIP	3.5.16	VISUAL AND		30			
DESIGN & CONSTRUCTION		DOCUMENTATION REVIEW		30			
GROUP II						0	30
TERMINAL STRENGTH	3.5.3	4.4.3.1		15			
SOLDERABILITY	3.5.5	4.4.3.2		15			
GROUP III						0	30
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3	10	10	10		
INSULATION RESISTANCE	3.5.9	4.4.3.4	10	10	10		
GROUP IV						0	300
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3					
ALL-FIRE (MN-AFV)	3.5.6.2	4.4.3.5	50	50			
WITH DENT TEST	3.5.6.1	4.4.3.6	50	50			
PROBIT ANALYSIS (VAF)	3.5.7	4.4.3.7	100				
PROBIT ANALYSIS (VNF)	3.5.7	4.4.3.7.2			100		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE I - (CONTINUED)

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	1/ °F + 5°F +75	1/ °F + 5°F +140	2/	3/
GROUP V						0	20
NO-FIRE CURRENT (MA-NFC)	3.5.6.3.2	4.4.3.8	5	5	10		
GROUP VI						0	20
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3		20			
SHOCK, THERMAL	3.5.10	4.4.3.9		20			
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3		20			
ALL-FIRE (MN-AFV)	3.5.6.2	4.4.3.5		20			
GROUP VII						0	20
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3		20			
SHOCK, MECHANICAL	3.5.12	4.4.3.10		20			
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3		20			
ALL-FIRE (MN-AFV)	3.5.6.2	4.4.3.5		20			
GROUP VIII						0	20
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3		20			
VIBRATION	3.5.13	4.4.3.11		20			
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3		20			
ALL-FIRE (MN-AFV)	3.5.6.2	4.4.3.5		20			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE I - (CONTINUED)

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	1/ °F + 5°F +75	1/ °F + 5°F +140	2/	3/
GROUP IX						0	20
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3		20			
MODIFIED T & H	3.5.11	4.4.3.12		20*			
LEAD TO LEAD RESISTANCE	3.5.8	4.4.3.3		20			
REDUCED PRESSURE AND ALL-FIRE (MN-AFV)	3.5.16 3.5.6.2	4.4.3.13 4.4.3.5		20			

1/ Testing temperature and test quantities required for the individual examination or test except as otherwise required in the test procedure paragraph. For exposure type tests, use temperature specified in test method.
2/ Number of defectives permitted in the test group.
3/ Total quantity of sample components required for test group.

* Temperatures as specified in MIL-STD-331.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.1.3 Fuze

4.3.3.1.3.1 The 12 fuzes submitted as part of the first article sample shall be tested in accordance with Paragraph 4.4.7. Should any item fail to meet the requirements, the first article sample shall be rejected.

4.3.3.1.3.2 Cover Assembly Pre-Potting Checkout. - The cover assemblies being fabricated as part of first article fuzes (see 4.3.3.1.3.1) shall be subjected to the testing defined in 3.8.15.4 prior to potting.

4.3.3.1.4 Battery, Reserve

4.3.3.1.4.1 First Article and Qualification Test. - The sample for this test shall be as specified in the following table. The battery shall be subjected to the examinations and tests in the order shown in Table 2. The total sample of batteries shall be selected and divided into the groups as specified and subjected to the inspections for their respective group. Test procedures will be as specified in the applicable paragraph as shown. Batteries used in this test will not be submitted as delivered quantities against the contract for intended use in final assembly fuzes. Items shall be tested at the temperature indicated unless otherwise specified in the procedural paragraph or referenced document. All the batteries submitted as the first article sample shall be subjected to a pre-initiation internal resistance test (see 3.7.20) - Major Defect - prior to any other testing.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the currentversion before use.

TABLE 2

BATTERY FIRST ARTICLE AND QUALIFICATION INSPECTION

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	1/ °F + 5°F +75	1/ °F + 5°F +140	2/	3/
GROUP I						0	14
MATCH RESISTANCE	3.7.6	4.4.5.6		14			
INSULATION RESISTANCE	3.7.7	4.4.5.7		14			
PRE-INITIATION VOLTAGE	3.7.8	4.4.5.8		14			
SEAL	3.7.3	4.4.5.2		14			
POLARITY	3.7.9	4.4.5.9			14		
NO FIRE	3.7.5.2	4.4.5.5		14			
GROUP II 4/						0	6
THERMAL SHOCK	3.7.12	4.4.5.12		6			
HUMIDITY	3.7.13	4.4.5.13			6		
TRANSPORTATION VIBRATION	3.7.14	4.4.5.14	3				
FLIGHT VIBRATION	3.7.15	4.4.5.15	3		3		
ALL-FIRE	3.7.15.1	4.4.5.4	3		3		
OUTPUT VOLTAGE	3.7.10	4.4.5.10	3		3		
NOISE VOLTAGE	3.7.11	4.4.5.11	3		3		
CASE CHANGE	3.7.1	4.4.5.18		6			
TERMINAL STRENGTH	3.7.2	4.4.5.1		6			

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 2 - (CONTINUED)

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	1/ °F + 5°F +75	1/ °F + 5°F +140	2/	3/
GROUP III 4/						0	8
THERMAL SHOCK	3.7.12	4.4.5.12		8			
HUMIDITY	3.7.13	4.4.5.13			8		
TRANSPORTATION VIBRATION	3.7.14	4.4.5.14	2	4	2	0	8
FLIGHT VIBRATION	3.7.15	4.4.5.15	3	3	2		
MECHANICAL SHOCK	3.7.17	4.4.5.17		8			
ACCELERATION	3.7.16	4.4.5.16		8			
ALL-FIRE	3.7.5.1	4.4.5.4		8			
OUTPUT VOLTAGE	3.7.10	4.4.5.10		8			
NOISE VOLTAGE	3.7.11	4.4.5.11		8			
CASE CHANGE	3.7.1	4.4.5.18		8			

1/ Testing temperature and test quantities required for the individual examination or test except as otherwise required by the test procedure paragraph. For exposure test, use temperature specified in test method.
2/ Number of defectives permitted in the test group.
3/ Total quantity of sample components required for test group.
4/ These samples obtained from Group I samples.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.1.5 Switch, Explosive, Multi-Circuit

4.3.3.1.5.1 First Article and Qualification Test. - The sample for this test shall be as specified in the following table. The switch shall be subjected to the examinations and tests in the order shown in Table 3. The total sample of switches shall be selected and divided into the groups as specified and subjected to the inspections for their respective group. Test procedures will be as specified in the applicable paragraph as shown in the table. Switches used in this test will not be submitted as delivered quantities against the contract for intended use in the final fuze assembly. Items will be tested at the temperatures indicated unless otherwise specified in the procedural paragraph or referenced document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 3

SWITCH, EXPLOSIVE, MULTI-CIRCUIT FIRST ARTICLE AND QUALIFICATION TEST

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	+75	+140	2/	3/
GROUP I						0	10
TERMINAL STRENGTH	3.8.1	4.4.6.1		10			
SOLDERABILITY	3.8.2	4.4.6.2		10			
GROUP II						0	10
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		10			
RESISTANCE, SWITCH	3.8.6	4.4.6.6		10			
INSULATION RESISTANCE	3.8.7	4.4.6.7	10				
ALL-FIRE	3.8.3	4.4.6.3	10				
RESISTANCE, SWITCH	3.8.6	4.4.6.6		10			
GROUP III						0	85
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		85			
SWITCH CONTINUITY	3.8.8	4.4.6.8		85			
NO-FIRE	3.8.4	4.4.6.4			85		
RESISTANCE, BRIDGEWIRE	3.8.6	4.4.6.6		85			
SWITCH CONTINUITY	3.8.8	4.4.6.8		85			
GROUP IV 4/						0	25
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		25			
SWITCH CONTINUITY	3.8.8	4.4.6.8		25			

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 3 - CONTINUED

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	1/ °F + 5°F +75	1/ °F + 5°F +140	2/	3/
GROUP IV - Con't							
SHOCK, THERMAL	3.8.9	4.4.6.9		25			
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		25			
ALL-FIRE	3.8.3	4.4.6.3	15	5	5		
SWITCH CONTINUITY	3.8.8	4.4.6.8		20			
GROUP V 4/						0	15
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		15			
SWITCH CONTINUITY	3.8.8	4.4.6.8		15			
SHOCK, MECHANICAL	3.8.10	4.4.6.10		15			
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		15			
SWITCH CONTINUITY	3.8.8	4.4.6.8		15			
ALL-FIRE	3.8.3	4.4.6.3	15				
SWITCH CONTINUITY	3.8.8	4.4.6.8		15			
GROUP VI 4/						0	15
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		15			
SWITCH CONTINUITY	3.8.8	4.4.6.8		15			
VIBRATION	3.8.11	4.4.6.11		15			
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		15			
SWITCH CONTINUITY	3.8.8	4.4.6.8		15			
ALL-FIRE	3.8.3	4.4.6.3			15		
SWITCH CONTINUITY	3.8.8	4.4.6.8		15			

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 3 - CONTINUED

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	+75	+140	2/	3/
GROUP VII 4/						0	10
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		10			
SWITCH CONTINUITY	3.8.8	4.4.6.8		10			
TEMPERATURE STORAGE	3.8.12	4.4.6.12		10			
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		10			
SWITCH CONTINUITY	3.8.8	4.4.6.8		10			
ALL-FIRE	3.8.3	4.4.6.3			10		
SWITCH CONTINUITY	3.8.8	4.4.6.8		10			
GROUP VIII 4/						0	20
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		20			
SWITCH CONTINUITY	3.8.8	4.4.6.8		20			
TEMP. AND HUMID.	3.8.13	4.4.6.13		20*			
RESISTANCE, BRIDGEWIRE	3.8.5	4.4.6.5		20			
SWITCH CONTINUITY	3.8.8	4.4.6.8		20			
ALL-FIRE	3.8.3	4.4.6.3	15		5		
SWITCH CONTINUITY	3.8.8	4.4.6.8		20			

1/ Testing temperature and test quantities required for the individual examination or test except as otherwise required in the test procedure paragraph. For exposure type test, use temperature specified in test method.
2/ Number of defectives permitted in the test group.
3/ Total quantity of sample components required for test group.
4/ Taken from Group III.
* Temperature as specified in MIL-STD-331.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.2 Production. - The components and assemblies submitted for acceptance shall be subjected to the test indicated below after having been examined for the requirements as specified in 4.3.2 where applicable.

4.3.3.2.1 Safety and Arming Device

4.3.3.2.1.1 Contact Resistance B-D and A-B (see 3.3.3) - Major Defect. - This test shall be performed 100 percent. Any assembly failing to meet the requirements shall be classed defective and removed from the lot. Testing shall be in accordance with 4.4.1.1.

4.3.3.2.1.2 Insulation Resistance (see 3.3.5) - Major Defect. - This test shall be performed 100 percent. Any Safety and Arming Device failing to meet the requirements shall be classed defective and removed from the lot. Testing shall be in accordance with 4.4.1.6.

4.3.3.2.1.3 Connector Pin to Pin Resistance (see 3.3.6) - Major Defect. - This test shall be performed 100 percent. Any Safety and Arming Device failing to meet the requirements shall be classed defective and removed from the lot. Testing shall be in accordance with 4.4.1.7.

4.3.3.2.1.4 Non-Arm Level (see 3.3.4) - Major Defect. - This test shall be performed 100 percent. Any Safety and Arming Device failing to meet the requirement shall be classed defective and removed from the lot. Testing shall be in accordance with 4.4.1.2.

4.3.3.2.1.5 Time to Arm (see 3.3.1) - Major Defect. - This test shall be performed 100 percent at cold temperature. This test shall be performed on a randomly selected sample of 80 units at hot temperature. The hot test shall be performed 100 percent if the lot size is less than 81 or if any unit fails to meet the requirement. Any unit failing to meet the requirement shall be classed defective and removed from the lot. Testing shall be in accordance with 4.4.1.3.

4.3.3.2.1.6 Setback Reset (see 3.3.2) - Major Defect. - This test shall be performed 100 percent. Any Safety and Arming Assembly failing to meet the requirements shall be classed defective and removed from the lot. Testing shall be in accordance with 4.4.1.5.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.2.1.7 Environmental Testing (see 3.3.7) - Major Defect. - This test shall be performed on 12 Safety and Arming Devices selected at random. The S&A's shall be mounted in fuze housings or containers that simulate the S&A cavity in the fuze housing. Failure of any unit to meet a part of the environmental requirement (Paragraph 3.3.7) as applicable, when tested immediately after exposure to the individual environment, shall be cause for rejection of the lot. Testing shall be in accordance with 4.4.1.8.2.

4.3.3.2.2 Piston Actuator

4.3.3.2.2.1 Initial Production Inspection of Piston Actuator. - The following tests shall be performed as specified in Table 4 which follows. Units for this testing shall be selected from the first production lot of components submitted. Any units found to be defective at any stage during testing shall be cause for rejection of the initial production sample. The tests specified within each group shall be performed in the order listed for each group. The groups may be tested in any order desired. This test shall be the criteria used for acceptance of the lot from which the units were drawn.

4.3.3.2.2.1.1 Lead to Lead Resistance (see 3.4.8) - Major Defect. - This test shall be performed on 100 percent of the lot. Testing shall be in accordance with 4.4.2.3. Any unit failing to meet the requirement shall be classed defective and removed from the lot.

4.3.3.2.2.1.2 Insulation Resistance (Pre-Fired) - (see 3.4.9.1) - Major Defect. - This test shall be performed on 100 percent of the lot. Testing shall be in accordance with 4.4.2.4. Any unit failing to meet the requirement shall be classed defective and removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 4

PISTON ACTUATOR INITIAL PRODUCTION INSPECTION

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	+75	+140	2/	3/
GROUP I						0	30
SOLDERABILITY	3.4.5	4.4.2.1		30			
TERMINAL STRENGTH	3.4.6	4.4.2.2		30			
GROUP II						0	30
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3	10	10	10		
INSULATION RESISTANCE (PRE-FIRE)	3.4.9.1	4.4.2.4	10	10	10		
GROUP III						0	120
PROBIT ANALYSIS ALL FIRE (VAF)	3.4.7	4.4.2.5	100				
ALL FIRE (MN-AFV) - MIN. ALL FIRE VOLTAGE	3.4.2	4.4.2.6		10	10		
GROUP IV						0	120
PROBIT ANALYSIS NO FIRE (VNF)	3.4.7	4.4.2.5			100		
NO-FIRE (MN-NFV) - MIN. NO-FIRE VOLTAGE	3.4.1.1	4.4.2.7	10	10			

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 4 - CONTINUED

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	1/ °F + 5°F +75	1/ °F + 5°F +140	2/	3/
GROUP V						0	20
NO FIRE (MA-NFC) - MAX. NO FIRE CURRENT	3.4.1.2	4.4.2.8	5	5	10		
GROUP VI						0	20
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		20			
INSULATION RESISTANCE (PRE-FIRE)	3.4.9.1	4.4.2.4		20			
SHOCK, THERMAL	3.4.10	4.4.2.10		20*			
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		20			
ALL FIRE (MN-AFV)	3.4.2	4.4.2.6		20			
INSULATION RESISTANCE (POST-FIRE)	3.9.4.2	4.4.2.4			20		
GROUP VII						0	20
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.5		20			
INSULATION RESISTANCE (PRE-FIRE)	3.4.9.1	4.4.2.4		20			
SHOCK, MECHANICAL	3.4.11	4.4.2.11		20*			
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		20			
ALL FIRE (MN-AFV)	3.4.2	4.4.2.6		20			
INSULATION RESISTANCE (POST-FIRE)	3.9.4.2	4.4.2.4			20		

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 4 - CONTINUED

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	+75	+140	2/	3/
GROUP VIII						0	20
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		20			
INSULATION RESISTANCE (PRE-FIRE)	3.4.9.1	4.4.2.4		20			
VIBRATION	3.4.12	4.4.2.12		20			
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		20			
ALL FIRE (MN-AFV)	3.4.2	4.4.2.6		20			
INSULATION RESISTANCE (POST-FIRE)	3.9.4.2	4.4.2.4			20		
GROUP IX						0	20
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		20			
INSULATION RESISTANCE (PRE-FIRE)	3.4.9.1	4.4.2.4		20			
MODIFIED TEMP & HUMID CYCLE	3.4.14	4.4.2.13		20*			
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		20			
NO FIRE (MA-NFC)	3.4.1.2	4.4.2.8		20			
INSULATION RESISTANCE (PRE-FIRE)	3.4.9.1	4.4.2.4		20			
REDUCED PRESSURE - ALL FIRE (MN-AFV)	3.4.2	4.4.2.14		20			

1/ Testing temperature and test quantities required for the individual examination or test except as otherwise required by test procedure paragraph. For exposure type test, use temperature specified in Test Method.
2/ Number of defectives permitted in the test group.
3/ Total quantity of sample components required for test group.
* Temperatures as specified in MIL-STD-331.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the currentversion before use.

4.3.3.2.2.2 Lot Acceptance Testing. - The following tests, as specified in Table 5, shall be performed on all production lots except those from which an initial production sample was drawn and accepted. The tests specified within each group shall be performed in the order listed for each group. The groups may be tested in any order desired. Any unit found defective at any stage of testing shall be cause for rejection of the lot from which they were selected.

4.3.3.2.2.2.1 Lead to Lead Resistance (see 3.4.8) - Major Defect. - This test shall be performed on 100 percent of the lot. Testing shall be in accordance with 4.4.2.3. Any unit failing to meet the requirement shall be classed defective and removed from the lot.

4.3.3.2.2.2.2 Insulation Resistance (Pre-Fired) - (see 3.4.9.1) - Major Defect. - This test shall be performed on 100 percent of the lot. Testing shall be in accordance with 4.4.2.4. Any unit failing to meet the requirement shall be classed defective and removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 5

PISTON ACTUATOR TABLE OF LOT ACCEPTANCE INSPECTION

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	1/ °F + 5°F +75	1/ °F + 5°F +140	2/	3/
GROUP I						0	10
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		10			
INSULATION RESISTANCE (PRE-FIRE)	3.4.9.1	4.4.2.4		10			
SHOCK, THERMAL	3.4.10	4.4.2.10		10			
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		10			
ALL FIRE (MA-FV)	3.4.2	4.4.2.6	10				
INSULATION RESISTANCE (POST-FIRE)	3.9.4.2	4.4.2.4			10		
GROUP II						0	50
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		50			
INSULATION RESISTANCE (PRE-FIRE)	3.4.9.1	4.4.2.4		50			
SHOCK, MECHANICAL	3.4.11	4.4.2.11		50			
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		50			
PROBIT ANALYSIS (ALL-FIRE)	3.4.7	4.4.2.5	50				
INSULATION RESISTANCE (POST-FIRE)	3.9.4.2	4.4.2.4			50		

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 5 - CONTINUED

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	1/ °F + 5°F +75	1/ °F + 5°F +140	2/	3/
GROUP III						0	10
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		10			
INSULATION RESISTANCE (PRE-FIRE)	3.4.9.1	4.4.2.4		10			
VIBRATION	3.4.12	4.4.2.12		10			
LEAD TO LEAD RESISTANCE	3.4.8	4.4.2.3		10			
ALL FIRE (MA-FV)	3.4.2	4.4.2.6	10				
INSULATION RESISTANCE (POST-FIRE)	3.9.4.2	4.4.2.4			10		

1/ Testing temperature and test quantities required for the individual examination or test except as otherwise required in the procedure paragraph. For exposure type test, use temperature specified in Test Method.
2/ Number of defectives permitted in the test group.
3/ Total quantity of sample components required for test group.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.2.3 Detonator

4.3.3.2.3.1 Each lot of detonators submitted for acceptance shall be subjected to the Qualification Test (see 4.3.3.1.2.1). Should any group fail to meet the requirements as stated, the lot shall be rejected.

4.3.3.2.3.2 Lead to Lead Resistance (see 3.5.8). - This test shall be performed on 100 percent of the items to be submitted for acceptance. If any item should fail to meet the requirement, it shall be removed from the lot.

4.3.3.2.3.3 Insulation Resistance (see 3.5.9). - This test shall be performed on 100 percent of the items to be incorporated into fuzes. If any item should fail to meet the requirement, it shall be removed from the lot. Test shall be in accordance with 4.4.3.4.

4.3.3.2.4 Fuze (see 3.6, 3.8.15 and 4.4.4). - Prior to the potting of the cover assembly, the testing of Paragraph 3.8.15.4 shall be performed. Each fuze, prior to final assembly of the three major subassemblies (i.e., cover [potted], housing and S&A) and during their assembly into the fuze shall be subjected to the following tests in the order listed. Any module (subassemblies, fuze less S&A or fuze) failing to meet the requirements of any test shall be cause for rejection of that module.

4.3.3.2.4.1 Vibration of Cover and Housing Assembly - Major Defect. - The cover and housing assembly mounted together shall be subjected to vibration over the frequency range of 5 to 500 HZ at a double amplitude of 0.5 inch or 3.5g's peak whichever is less. Sweep time from 5 to 500 to 5 HZ shall be 15 minutes and shall be a logrithmic sweep. The total duration of vibration shall be 15 minutes in each of three mutually perpendicular axis. After the test, the housing and cover shall be disassembled and subjected to the following tests. The housing assembly shall be tested to meet requirements of 3.6.5.2, 3.6.5.3 and 3.6.5.4. The cover assembly shall be tested to meet requirements of 3.8.15.1, 3.8.15.2 and 3.8.15.3.

4.3.3.2.4.2 Fuze Less S&A Assembly Test - Major Defect. - The fuze, less S&A assembly, shall be subjected to the timing requirements of 3.6.5.1 and 4.4.4.4 in accordance with the following specified condition and voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

CONDITION	VOLTAGE	TEST METHOD
Ambient	6 Volts DC	4.4.4.4
- 40°F	5 Volts DC	4.4.4.4
+140°F	9 Volts DC	4.4.4.4
Random Vibration at Ambient	6 Volts DC	4.4.4.3
Ambient	6 Volts DC	4.4.4.4

4.3.3.2.4.3 Setback Reset - Major Defect. - This test shall be performed according to Paragraph 4.4.4.1.

4.3.3.2.4.4 Resistance Test of Fuze - Major Defect. - Each fuze shall be tested to the requirements of Paragraph 3.6.6.1 and 3.6.6.2.

4.3.3.5 Battery, Reserve, Primary

4.3.3.5.1 Each lot of batteries submitted for acceptance shall be subjected to the lot acceptance tests specified in Table 6. The tests specified within each group shall be performed in the order listed. The groups may be tested in any order desired. Batteries subjected to Group I tests may be used for in-fuze assembly. Group II batteries shall not be used in fuze assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

TABLE 6

BATTERY, RESERVE, PRIMARY, ACCEPTANCE INSPECTIONS FOR

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	1/ °F + 5°F -40	+75	+140	2/	3/
GROUP I						0	100%
BATTERY PREINITIATION INTERNAL RESISTANCE	3.7.20	4.4.5.19		100%			
SEAL	3.7.3	4.4.5.2		100%			
MATCH RESISTANCE	3.7.6	4.4.5.6		100%			
POLARITY	3.7.9	4.4.5.9			100%		
INSULATION RESISTANCE	3.3.7	4.4.5.7		100%			
PREINITIATION VOLTAGE	3.7.8	4.4.5.8		100%			

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	AQL, PERCENT DEFECTIVE	SAMPLING PLAN (MIL-STD-105)
GROUP II				
NO FIRE	3.7.5.2	4.4.5.5	1.5	
ALL FIRE	3.7.5.1	4.4.5.4		LEVEL
OUTPUT VOLTAGE	3.7.10	4.4.5.10		S4
NOISE VOLTAGE	3.7.11	4.4.5.11		
CASE CHANGE	3.7.1	4.		TABLE I

1/ Test Temperature - For Group II - Test one half of sample at -45°F + 5°F and one half at +145°F.
2/ Allowable defectives permitted.
3/ Number inspected.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.3.3.2.6 Switch, Explosive, Multi-Circuit

4.3.3.2.6.1 Each lot of switches submitted for acceptance shall be subjected to the Qualification Test specified in 4.3.3.1.5. Should any group fail to meet the requirement as stated, the lot shall be rejected.

4.3.3.2.6.2 Input (Bridgewire) Resistance - Major Defect. - This test shall be performed on 100 percent of the switches submitted for acceptance. If any item should fail to meet the requirement, it shall be removed from the lot.

4.3.3.2.6.3 Pre-Fired Continuity - Major Defect. - This test shall be performed on 100 percent of the switches submitted for acceptance. If any item should fail to meet the requirement, it shall be removed from the lot.

4.3.3.2.7 Rotary Switch Test - Major Defect. - This test shall be performed 100 percent on all rotary switches. If any item should fail to meet the requirements of Paragraph 3.10 and 4.4.8, it shall be removed from the lot and rejected.

4.3.3.2.8 Tuning Fork Resonator (dwg. 9251908) - Major Defect. - Prior to soldering the cover, a 100% visual inspection shall be performed to assure that the wires are in the proper position, adhesives in the proper location and good workmanship per 9251908. After sealing the tests and inspection as specified on dwg. 9208806 shall be performed.

4.3.3.2.9 Rotary Switch (dwg. 9208789) - Major Defect. - The following visual inspections shall be performed 100% during the assembly of the Rotary Switch (dwg. 9208789). Inspect each shaft-rotor assembly after the flare operation for cracks in the molded rotor or damaged roll-over. Inspect each housing detent assembly using 20X microscope for no physical damage, no foreign particles and lubricants on or in area of rotor. Immediately after the wipers are inserted into the slots on the rotor, a visual inspection shall be performed to insure that wipers are mounted correctly.

4.3.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed in 4.3.2 is identified under the "Method of Inspection" headings, and unless otherwise specified, shall be contractor designed. Unless otherwise specified, the inspection equipment used to perform the examinations and tests prescribed in 4.3.3 shall be contractor design.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.4 Testing Procedures and Methods

4.4.1 Safety and Arming Device

4.4.1.1 Contact Resistance (see 3.3.3.2). - This test shall be performed in the unarmed rotor position. The detonators shall both be shorted. The test currents shall be limited to one (1) milliampere maximum. A mating connector shall be plugged into the unit and resistance measured across the lead wires which shall be the same type as those used in the fuze.

4.4.1.2 Non-Arming Level (see 3.3.4). - The device shall be subjected to a steady state acceleration of 8.8g's + .2g's for a minimum of ten (10) seconds. The force shall be applied perpendicular to the base in the direction that tends to arm the device. The detonators shall be shorted during this test. Test current shall be limited to one (1) milliampere maximum.

4.4.1.3 Time to Arm (see 3.3.1). - This test shall be performed with detonators shorted. The device shall be subjected to a step input of 14g's perpendicular to the base in the direction which tends to arm the device with a rise time on the input to be less than 120 milliseconds. Test current shall be limited to one (1) milliampere maximum. An electrical timer shall be used to measure contact break to contact make time.

4.4.1.4 Contact Resistance (see 3.3.3.1). - This test shall be performed in the armed rotor position. The detonators shall be shorted. The test currents shall be limited to one (1) milliampere maximum. A mating connector shall be plugged into the unit and resistance measured across the leadwires which shall be no longer than two inches and of the same type as in the fuze.

4.4.1.5 Setback Reset (see 3.3.2). - The setback weight of the S&A Device shall be checked for freedom of vertical movement and reset capability. The weight shall be depressed by the application of a one pound (max.) force, exerted by a blunt probe between the rotors acting upon the setback weight. The weight shall exhibit a full vertical movement as defined by a condition in which the rotors are manually rotated to the armed position and exhibit a self locking capability. The rotors, after having been placed in the armed locked condition, shall be unlocked and the reset capability of the mechanism tested. Compliance with 3.3.2 shall be ascertained.

4.4.1.6 Insulation Resistance (see 3.3.5). - Test currents shall be limited to one (1) milliampere. Voltage shall be 50 ± 1 Volt D.C. The meter shall be placed across the appropriate

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

leads of the mating connector and the resistance measured and recorded. There shall be no shorts on the detonators during this test. The mating connector leads shall be of the same type wire as the fuze leads.

4.4.1.7 Connector Pin to Pin Resistance (see 3.3.6). - Test currents shall be limited to one (1) milliampere maximum. The meter shall bt placed across the appropriate terminals through the use of a mating connector and the resistance measured and recorded. Detonators shall not be shorted during this test. The mating connector leads shall be of the same type wire as the fuze leads.

4.4.1.8 Environmental Testing (see 3.3.7)

4.4.1.8.1 First Article

4.4.1.8.1.1 The 24 units shall be consecutively numbered and subjected to the following environments as applicable and in the order listed.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

UNIT NUMBERS	(1,2,3)	(4,5,6)	(7,8,9)	(10,11,12)	(13,14,15)	(16,17,18)	(19,20,21)	(22,23,24)
ENVIRONMENTS								
Transportation Vibration	No	Yes	Yes	No	Yes	No	No	Yes
Thermal Shock	No	No	Yes	Yes	No	Yes	Yes	Yes

SEQUENCE

All Above Units
at -40 Degrees F - 3F°

All Above Units
at +140 Degrees F + 3F°

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

Transportation Vibration. - The units shall be tested as specified in MIL-STD-810B with the following exceptions. The units shall be subjected to vibration over the frequency range of 5 to 500 HZ at a double amplitude of 0.5 inch or 3.5g's peak, whichever is less. Sweep time from 5 to 500 to 5 HZ shall be 15 minutes and shall be a logrithmic sweep. The total duration of vibration shall be four (4) hours in each of the three mutually perpendicular axis. The units to be tested cold shall be conditioned at -65°F and the units to be tested hot shall be conditioned at +160°F.

Thermal Shock. - In accordance with MIL-STD-331, Test Method 113. Sequence as follows:

Non-Arming Level in accordance with 4.4.1.2 with the exception that the detonator leads shall NOT be shorted.

Time to Arm in accordance with 4.4.1.3 with the exception that the detonator leads shall NOT be shorted.

Missile Vibration

Random three (3) axes, 0.06g /CPS, 100-1000 CPS attenuate both ends at 12 db/octave, 9.3 grms overall, 10 second duration.

Random three (3) axes, 0.02g /CPS, 100-1000 CPS attenuate both ends at 12 db/octave, 5.3 grms overall, 120 seconds duration.

The piston actuators shall be initiated using a simulated firing circuit shown in Figure 2 with the test voltage to be per Paragraph 3.4.2.

The detonators shall be initiated using a simulated firing circuit shown in Figure 2 with the test voltage to be per Paragraph 3.5.6.2. Functioning shall be evidenced by an open in the monitor loop (Pins C&E) and evidence of lead initiation as shown by one or two holes blown through the bottom of the S&A base plate. The cause of any malfunction shall be determined radiographically.

4.4.1.8.1.1 Jolt Test. - The units shall be subjected to the jolt test of MIL-STD-331, Test Method 101. After disassembly, it shall be visually determined that neither the detonators, leads or piston activators have functioned.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.1.8.2 Production. - The 12 units selected shall be subjected to the environmental sequence listed below.

ENVIRONMENTAL SEQUENCE	UNIT NUMBER					
	1,2	3,4	5,6	7,8	9,10	11,12
Transportation Vibration (Amb)	X	X	X	X	X	X
Thermal Shock			X	X	X	X
Non-Arming (- 40°F)	X		X		X	
Non-Arming (+140°F)		X		X		X
Time to Arm (- 40°F)	X		X		X	
Time to Arm (+140°F)		X		X		X
Missile Vibration (- 40°F)	X		X		X	
Missile Vibration (+140°F)		X		X		X
Lead Arming (- 40°F)	X		X		X	
Lead Arming (+140°F)		X		X		X
Detonator Function (- 40°F)	X		X		X	
Detonator Function (+140°F)		X		X		X

The environments shall be as defined in 4.4.1.8.1.1 with the exception that Transportation Vibration shall be done at ambient temperature.

4.4.2 Piston Actuator

4.4.2.1 Solderability (see 3.4.5). - This test can be performed on units subjected to the all fire test (see 4.4.2.6). There shall be no special preparation of terminals prior to testing. Testing shall be in accordance with MIL-STD-202, Test Method 208.

4.4.2.2 Terminal Strength (see 3.4.6). - The piston actuator shall be tested in accordance with MIL-STD-202, Test Method 211, Test Condition A except that the applied force shall be three (3) pounds minimum for a period of 30 seconds minimum.

4.4.2.3 Lead to Lead Resistance (see 3.4.8). - This test shall be performed in accordance with MIL-STD-202, Test Method 303 using a current of 1.1 milliamperes maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.2.4 Insulation Resistance, Pre-Fired Condition (see 3.4.9.1) - Post Fired Condition (see 3.4.9.2). - This test shall be performed in accordance with MIL-STD-202, Test Method 302. The test potential shall be 50 volts DC ± 5 volts applied between the lead wires shunted together and the case.

4.4.2.5 Probit Analysis (see 3.4.7). - The minimum "all fire" voltage and maximum "no fire" voltage shall be determined using capacitor discharge testing. They shall be determined through the use of probit testing as specified in NAVORD Report 2101 using the input voltage "V" as specified in Figure 2 as the independent variable. The piston actuator output force and operating time (see 3.4.3) shall be measured and used as the criteria for success or failure of the individual operation. The testing will be performed at not less than five voltages with a minimum sample of 20 units at each voltage and at the temperature as indicated by Table 4, Group III and Group IV.

4.4.2.5.1 "All Fire". - The minimum "all fire" voltage shall be computed from the test data as follows:

$$VAF = \bar{V}UCL + 3\ UCL$$

$\bar{V}UCL$ = Mean firing voltage at the upper 60% confidence limit.
UCL = Standard deviation of firing voltage at the upper 60% confidence limit.

4.4.2.5.2 "No Fire". - The maximum "no fire" voltage shall be computed from the test data as follows:

$$VNF = \bar{V}LCL - 4\ LCL$$

$\bar{V}LCL$ = Mean firing voltage at the lower 60% confidence limit.
LCL = Standard deviation of firing voltage at the upper 60% confidence limit.

4.4.2.6 Minimum "All Fire" Voltage (see 3.4.2). - This test shall be performed in accordance with Figure 2. Input voltage shall be 5.0 to 4.9 VDC. The SCR shall be fired by an external force.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.2.7 Maximum "No Fire" Voltage (see 3.4.1.1). - This test shall be performed in accordance with Figure 2 with the exception that the input voltage shall be 2.0 to 2.5 VDC.

4.4.2.8 Maximum "No Fire" Current (see 3.4.1.2). - The component shall be tested by applying 100 + 5 milliamperes DC through its lead wires (unshunted) for a minimum of three minutes.

4.4.2.9 Lead to Lead Resistance (see 3.4.1.2). - This test shall be performed in accordance with MIL-STD-202, Test Method 303 with a test current of 1.1 milliamperes maximum.

4.4.2.10 Thermal Shock (see 3.4.10). - The piston actuator shall be tested in accordance with MIL-STD-202, Test Method 107, Test Condition A with the exceptions as follows:

a. The high temperature shall be +162°F ± 5F° in lieu of 85°F + 3°C.

b. The exposure duration at each temperature shall be four (4) hours minimum.

All measurements called for shall be performed only after completion of Environmental Exposure, as indicated herein.

4.4.2.11 Mechanical Shock (see 3.4.11). - This test shall be performed in accordance with MIL-STD-202, Test Method 213. The piston actuator shall be rigidly mounted in a manner which permits testing after environmental exposure test. Condition F will be used. The shock pulse shall approximate a half-sine wave. Each actuator shall receive three pulses, one each in the following axis:

a. Longitudinal axis of the actuator vertical with the leads up.

b. Longitudinal axis of the actuator vertical with the leads down.

c. Longitudinal axis of the actuator horizontal.

All measurements called for shall be performed only at completion of Environmental Exposure, as indicated herein.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.2.12 High Frequency Vibration (see 3.4.12). - This test shall be performed in accordance with MIL-STD-202, Test Method 204B, Test Condition D. The actuator shall be rigidly mounted to the test equipment and the lead wires secured. The frequency range shall be 10 to 2000 HZ + 10% with a vibration amplitude of .4 inch DA or 30g's whichever is less. Cycling rate shall be 20 minutes + 2 from 10 to 2000 HZ and back to 10 HZ. The vibration time shall be four hours in each of three mutually perpendicular axes (12 hours total). All measurements called for shall be performed only after completion of all environmental exposure, as indicated herein.

4.4.2.13 Modified Temperature and Humidity Cycle (see 3.4.14). - The piston actuator shall be subjected to a modified T&H cycle of 12 day duration in accordance with MIL-STD-331, Test Method 105. The test shall be in accordance with the cycle specified for the first 12 days of Method 105 of MIL-STD-331.

4.4.2.14 Reduced Pressure (see 3.4.2). - This test shall be performed at -40°F + 5F° in accordance with MIL-STD-202, Test Method 105C, Condition C. While at these conditions, the all fire test called for in Group IX of Table 1 shall be performed in accordance with 4.4.2.6.

4.4.2.15 Exhibited Force (see 3.4.3). - The piston shall be mounted perpendicular to the direction in which gravity would aid passing of the test.

4.4.3 Detonator

4.4.3.1 Terminal Strength (see 3.5.3)

4.4.3.1.1 Pull Test. - The detonator shall be tested in accordance with MIL-STD-202, Test Method 211, Test Condition A except that the applied force shall be three (3) pounds minimum for a period of 30 seconds minimum.

4.4.3.1.2 Bend Test. - The detonator shall be tested in accordance with MIL-STD-202, Test Method 211, Test Condition C using a two and one half (2.5) pound force.

4.4.3.2 Solderabilityy (see 3.5.5). - The detonator shall be tested in accordance with MIL-STD-202, Test Method 208. There shall be no special preparation of the terminals.

4.4.3.3 Lead to Lead Resistance (see 3.5.8). - The DC resistance across the bridgewire shall be measured in accordance with MIL-STD-202, Test Method 303 using a current of 1.1 milliamperes maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.3.4 Insulation Resistance (see 3.5.9). - The detonator shall be tested in accordance with MIL-STD-202, Test Method 302. The test potential shall be applied between the shorted lead wires and the case. The test voltage shall be 50 VDC ± 5 VDC minimum.

4.4.3.5 Minimum All Fire Voltage (MN-AFV) - (see 3.5.6.2). - The detonator shall be initiated using the circuitry shown in Figure 2. The SCR shall be fired by an external source and the operating time of the detonator measured. A component which does not initiate in the first attempt shall be considered a failure. Firing voltage shall be 5.0 to 4.9 VDC.

4.4.3.6 Dent Test (see 3.5.6.1). - The detonator shall be installed in the test fixture as depicted in Figure 3. Initiation shall be by the use of the circuit specified in 4.4.3.5. The indentation(s) made in the witness plate(s) shall be measured with a dial indicator height gage accurate to 0.0005 inch. The dial indicator shall be equipped with a probe conforming to Figure 4. Witness plate(s) shall be made from 1018 to 1020 steel having a Rockwell Hardness Number of B70 to B95 as measured on the dent test surface. The surface on all sides used for testing shall have RMS finish of 63 microinches maximum as measured with a profilemeter.

4.4.3.7 Probit Analysis (see 3.5.7). - The minimum "all fire" voltage and maximum "no fire" voltage shall be determined using capacitor discharge testing. They shall be determined through the use of probit testing, as specified in NAVORD Report 2101 using the input voltage "V" as specified in Figure 2 as the independent variable. The testing will be performed at not less than five voltages with a minimum sample of 20 units at each voltage and at the temperature indicated by Table 1.

4.4.3.7.1 "All Fire". - The minimum "all fire" voltage shall be computed from the test data as follows:

$$VAF = \overline{V}UCL + 3\ \sigma UCL$$

$\overline{V}UCL$ = Mean firing voltage at the upper 60% confidence limit.
σUCL = Standard deviation of firing voltage at the upper 60% confidence limit.

4.4.3.7.2 "No Fire". - The maximum "no fire" voltage shall be computed from the test data as follows:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

$$VNF = \bar{V}LCL - 4\ \sigma LCL$$

$\bar{V}LCL$ = Mean firing voltage at the lower 60% confidence limit.
σLCL = Standard deviation of firing voltage at the lower 60% confidence limit.

4.4.3.8 Maximum No Fire Current (MA-NFC) - (see 3.5.6.3.2).- The detonator shall be subjected to a 50 + 0/-2 milliampere current applied through its unshunted lead wires for a duration of five (5) minutes minimum.

4.4.3.9 Thermal Shock (see 3.5.10). - The detonator shall be tested in accordance with MIL-STD-202, Test Method 107C, Test Condition A with the exceptions as follows:

a. The high temperature shall be +162°F ± 5F° in lieu of 85°C + 3°C.

b. The exposure duration at each temperature shall be four (4) hours minimum.

All measurements called for shall be performed only after completion of environmental exposure, as indicated herein.

4.4.3.10 Mechanical Shock (see 3.5.12). - This test shall be performed in accordance with MIL-STD-202, Test Method 213. The detonator shall be rigidly mounted in a manner which permits testing after environmental exposure. Test Condition F will be used. Ea ch detonator shall receive six (6) pulses, two each in the following orientations:

a. Longitudinal axis of the detonator vertical with the leads up.
b. Longitudinal axis of the detonator vertical with the leads down.
c. Longitudinal axis of the detonator horizontal.

4.4.3.11 Vibration (see 3.5.13). - The detonator shall be tested in accordance with MIL-STD-810, Method 514, Procedure I, Curve H, Parts 1, 2 and 3.

4.4.3.12 Modified Temperature and Humidity Cycle (see 3.5.11). - The detonator shall be subjected to a modified T&H cycle of 12 day duration in accordance with MIL-STD-331, Test Method 105.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.3.13 Reduced Pressure (see 3.5.16). - This test shall be performed at -40°F + 5F° in accordance with MIL-STD-202, Test Method 105C, Condition C. While at these conditions, the all fire test called for in Group IX of Table 1 shall be performed in accordance with 4.4.3.5.

4.4.4 Fuze, XM811E6 Pre-Assembly and Checkout (see 3.6). - The following examinations and tests shall be performed on an S&A Device, Cover Assembly and Housing Assembly, as applicable, intended to be finally mated together during this examination into the final product. Testing shall be in the order indicated below.

4.4.4.1 Setback Reset (see 3.3.2). - The setback weight of the S&A Device shall be checked for freedom of vertical movement and reset capability. The weight shall be depressed by the application of a one pound maximum force, exerted by a blunt probe between the rotors, acting upon the setback weight. The weight shall exhibit a full vertical movement as defined by a condition in which the rotors are manually rotated to the armed position and exhibit a self locking capability. The rotors, after having been placed in the armed locked condition, shall be unlocked and the reset capability of the mechanism tested. Compliance with 3.3.2 shall be ascertained.

4.4.4.2 DELETED

4.4.4.3 The Device shall meet the requirements of Paragraph 4.4.4.4 while being subjected to vibration per Figure 7 in the lateral axis. The Device shall meet the same requirements in the vertical and longitudinal axis except that only times of event 199.9, SCO J and Arm 80 shall be used.

4.4.4.4 SECO, Arm, Event Test. - The fuze, less S&A, shall be electrically connected to the voltage specified in Paragraph 4.4.4.4 and the following combination of SECO, Arm and Event times set and measured. The maximum current that the Device shall draw during this test shall be 600 milliamps.

SECO	ARM	EVENT
A	10	*011.1
B	20	**022.2
C	**10	***033.3
D	40	*044.4
E	***10	**055.5
F	80	***066.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

SECO	ARM	EVENT
G	80	*077.7
H	80	**088.8
J	80	***099.9
OFF	80	*100.0
OFF	80	**100.0
OFF	80	***100.0

*, **, *** - There are three "zero" settings and three "one" settings on the first Event Dial. They shall all be checked as indicated.

4.4.4.5 DELETED

4.4.4.6 DELETED

4.4.5 Battery, Reserve

4.4.5.1 Terminal Strength (see 3.7.2). - The battery shall be tested in accordance with MIL-STD-202, Test Method 211, Test Condition A except that the applied force shall be two (2.0) pounds minimum for a period of 30 seconds minimum.

4.4.5.2 Seal (see 3.7.3). - The battery shall be tested in accordance with MIL-STD-202, Test Method 112, Test Condition C, Procedure IIIa. The following details will apply.

a. Backfilling Pressure: 70 ± 5 PSIG for two hours minimum.
b. Pressure Differential: One (1) atmosphere.
c. Duration of Test: Five (5) minutes.

4.4.5.3 DELETED

4.4.5.4 All Fire (see 3.7.5.1). - The battery shall be tested for initiation by applying 3.5 ± 0.1 amperes DC through the match lead wires for 22.5 ± 2.5 milliseconds maximum or until the bridgewire opens, whichever is less. Initiation shall be evidenced by an increase of the component output voltage from 0.15 DC maximum initial to 3.8 VDC minimum final.

4.4.5.5 No Fire (see 3.7.5.2). - The battery shall be tested by applying 1.0 ± 0.1 amperes DC through the match (bridgewire) lead wires for 4.0 ± 1.0 seconds. Initiation shall be evidenced by an increase of the component output voltage by 0.5 VDC minimum from the original 0.15 VDC maximum.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.5.6 Match Resistance (see 3.7.6). - The battery shall be tested in accordance with MIL-STD-202, Test Method 303. The test current shall be limited to 50 milliamperes DC maximum. The measurement shall be taken between lead wires. The meter shall be connected 1.5 inches in distance from the case along each lead wire.

4.4.5.7 Insulation Resistance (see 3.7.7). - The battery shall be tested in accordance with MIL-STD-202, Test Method 302, Test Condition A. The points of measurement shall be between:

a. The output terminals shorted together and the case.
b. The output terminals shorted together and the match lead wires shorted together.
c. The match lead wires shorted together and the case.

4.4.5.8 Pre-Initiation Voltage (see 3.7.8). - The battery shall be tested with a voltmeter having an input impedance of 10 megohms ± 20 percent. The measurement shall be taken between output terminals.

4.4.5.9 Polarity (see 3.7.9). - The battery shall be tested with a voltmeter having an input impedance of 10 megohms ± 20 percent. The measurement shall be taken from output terminal (-) to output terminal (+).

4.4.5.10 Output Voltage (see 3.7.10). - The output of the battery shall be measured with a 14.0 ± 0.1 ohm load connected across the output terminals. (This includes the input impedance of the measuring circuit.) The component shall be initiated as specified in 4.4.5.4. The following measurements shall be taken:

a. Pre-Initiation Match (Bridgewire) Resistance (see 4.4.5.6)

b. Time of application of initiation current pulse.

c. Component output voltage.

1. Elapsed time from initiation for output voltage to exceed 6.1 VDC and to exceed 6.5 VDC.

2. Elapsed time from initiation for output voltage to decrease to less than 6.5 VDC.

3. Maximum output voltage level.

4. Noise voltage (see 4.4.5.11).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

NOTE: Failure to comply with this test requirement due to a failure in a. above shall be reported as a failure of 3.7.6 and, due to 4. above, shall be reported as a failure of 3.7.11.

4.4.5.11 Noise Voltage (see 3.7.11). - Spurious changes in the battery output voltage shall be measured during the active life of the battery (6.1 VDC min. to 8.3 VDC max. to 6.5 VDC min.). The test shall be performed in conjunction with the output voltage test of 4.4.5.10. The measuring instrumentation shall have a band width of 80 to 2000 HERTZ.

4.4.5.12 Thermal Shock (see 3.7.12). - The batteries shall be tested in accordance with MIL-STD-331, Test Method 113. No measurements shall be taken during this exposure. Test temperature shall be -40°F to +140°F.

4.4.5.13 Humidity (see 3.7.13). - The battery shall be tested as specified in MIL-STD-202, Test Method 103, Test Condition A. Test temperatures shall be 140 + 5°F. Relative humidity shall be 95 ± 2%. No measurements shall be taken during this exposure.

4.4.5.14 DELETED

4.4.5.15 Flight Vibration (see 3.7.15). - The battery shall be tested in accordance with MIL-STD-810, Test Method 514, Procedure II, Part 3, Curve AG. The following details and exceptions apply.

a. Test temperatures will be as specified in the Qualification Inspection Table (#2).

b. Each battery will be tested in three (3) mutially perpendicular axes, one of which is the cylindrical axis.

c. Duration of test vibration shall be five (5) minutes per axis.

d. When vibration environment is specified in the inspection tables, prior to all fire inspections, approximately equal quantities will be initiated and operated in each axis. The battery shall be initiated prior to vibration in the third axis and vibration shall start within five seconds thereafter.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.5.16 Acceleration (see 3.7.16). - The battery shall be tested as specified in MIL-STD-810, Method 513, Procedure II. The following additions and exceptions apply.

a. The acceleration levels shall be 4.0 ± 0.5g along the longitudinal (cylindrical) axes and 40 ± 1g normal to the longitudinal (cylindrical) axes.

b. Duration of the test shall be one minute in each orientation minimum, except during the operational test which shall be five minutes minimum.

c. Unit orientation for the operational test shall be approximately equal. One third of the sample shall be along the cylindrical axes, on e third in the opposite direction along the same axis and one third normal to the cylindrical axis.

4.4.5.17 Mechanical Shock (see 3.7.17). - The batteries shall be tested as specified in MIL-STD-202, Method 213, Condition A.

4.4.5.18 Case Change (see 3.7.1). - The case shall be mechanically measured for compliance to the requirement.

4.4.5.19 Battery Pre-Initiation Internal Resistance. The internal resistance of the battery shall be measured across the positive and negative terminals of the battery under a test voltage of no greater than 50 VDC under a maximum current of one (1) milliampere.

4.4.6 Switch, Explosive, Multi-Circuit

4.4.6.1 Terminal Strength (see 3.8.1). - The switch shall be tested in accordance with MIL-STD-202, Test Method 211, Test Condition A with the exception that the applied force shall be three (3) pounds for a period of 30 seconds minimum.

4.4.6.2 Solderability (see 3.8.2). - The switch shall be subjected to this test in accordance with MIL-STD-202, Test Method 208. There shall be no special preparation of the terminals or lead wires.

4.4.6.3 All-Fire Characteristics (see 3.8.3). - The switch shall be initiated by firing the SCR (see Figure 2) with an external source. Component operating time shall be measured. A voltage of 5.0 to 4.9 shall be used.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the currentversion before use.

4.4.6.4 No-Fire Characteristics (see 3.8.4). - The switch shall be subjected to a constant DC current of 40 milliamperes through its unshunted leadwires for a period of five (5) minutes minimum.

4.4.6.5 Input (Bridgewire) Resistance (see 3.8.5). - The input leadwire to leadwire resistance shall be measured in accordance with MIL-STD-202, Test Method 303 using a test current of 1.0 ± 0.1 milliampere.

4.4.6.7 Insulation Resistance (see 3.8.7). - The switch shall be tested in accordance with MIL-STD-202, Test Method 302 with the exception that the test potential shall be 500 ± 10 VDC. The points of application shall be as follows:

a. Each termainl to the case.

b. Each pair of switch terminals shorted together to every other pair of switch terminals.

c. All switch termainls electrically shorted together to the input (bridgewire). Terminals electrically shorted together.

4.4.6.8 Resistance (see 3.8.8). - The resistance shall be measured by placing the meter across the applicable terminals.

4.4.6.9 Thermal Shock (see 3.8.9). - The switch shall be tested as specified in MIL-STD-202, Test Method 107, Test Condition A using a high temperature of +160°F + 5F°. Exposure duration at each temperature extreme shall be four (4) hours minimum. Each component of the sample shall be subjected to the tests specified for its group following the last four (4) hour cycle and while still exposed to its respective temperature. Components to be tested at 75°F shall be stabilized at that temperature for four (4) hours following the last cycle of thermal shock before testing.

4.4.6.10 Mechanical Shock (see 3.8.10). - The switches shall be subjected to mechanical shock as specified in MIL-STD-202, Test Method 213, Test Condition F. The components shall be rigidly mounted and subjected to one shock in each direction of the three major axes. During the shock test, the switch (output) circuit shall be monitored for contact chatter.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

4.4.6.11 Vibration (see 3.8.11). - The switch shall be tested in accordance with MIL-STD-810, Test Method 514, Procedure I, Curve H, Parts 1, 2 and 3. Tests following vibration in the inspection table shall be performed during the vibration environment. The switch chatter shall be monitored during performance of the continuity check out and shall not exceed the requirement. Both normally closed pre-fired and post-fired sets of contacts shall be monitored for chatter.

4.4.6.12 Temperature Storage (see 3.8.12). - The switch shall be placed in a temperature chamber for 28 ± 2 days at +160°F ± 5F° with a relative humidity of less than 20 percent followed by exposure to -65F° + 5F° for an additional 28 ± 2 days. The following details shall apply:

a. Transfer from one temperature to the other may be accomplished in a single chamber at the rate of 100°F ± 20F° per hour or separate chambers may be used with the component conditioned at 72°F ± 10F° for one hour minimum during transfer.

b. The air in the chamber(s) shall be moved at the rate of 20 ± 5 times the chamber volume per minute when measured at 75°F ± 5F°.

c. The total volume of components in the chamber(s) shall not exceed 1/3 the volume of the chamber(s).

d. The component shall be mounted in the chamber(s) such that there is substantially no obstruction to the free flow of air around the component.

4.4.6.13 Temperature and Humidity Cycle (see 3.8.13). - The switch shall be subjected to the T&H cycle in accordance with MIL-STD-331, Test Method 105.

4.4.7 Fuze Environmental Checkout (see 3.9). - The 12 fuzes shall be numbered 1 through 12 inclusive. The S&A's shall be removed from all fuzes with the exception of Numbers 4 and 5. Ten (10) replacement S&A simulators consisting of base, piston actuator assembly, two (2) MMS switches to simulate the detonators and an appropriate circuitry to allow fuze T.M. of SECO and Event on Pin 11 of the J1 connector. The 10 simulators shall be secured in the same manner as the S&A's which they replace. The detonator rotors of the remaining two fuzes shall be armed in a 14g field. Maximum duration of the g field shall be 10 seconds. The detonator rotor arming shall be accompoished after exposure to the non-operating

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

environment and prior to operating environment. The 12 fuzes shall be subjected to the following matrix environments and appropriate time outs as listed below. Time outs shall be made while under the operating environment after previous exposure to the non-operating environment. For the temperature and humidity tests, the fuze shall be mounted to a Warhead Fuze Ring (dwg. 9283969) with the Fuze Gasket (dwg. 9267741) in place, together with Cable Assembly, XM251 (dwg. 9208657) and Gasket Fuze Ring Adapter (dwg. 9208676).

NOTE: Units 4 and 5 will have explosive output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

ENVIRONMENTAL MATRIX

OPERATING/ NON-OPERATING	TIME OUT UNDER RANDOM VIBRATION -40°F	TIME OUT UNDER RANDOM VIBRATION +140°F	TIME OUT UNDER R.F. ENVIRONMENT AMBIENT
No Environment	Fuze 1 R.V. Axis 1	Fuze 2 R.V. Axis 2	Fuze 3
Temp Humid Shock (T&H w/TS Transfer Rate)	Fuze 4 R.V. Axis 2 S&A	Fuze 5 R.V. Axis 3 S&A	Fuze 6
T.V. +160 and M.S. +160	Fuze 7 R.V. Axis 3	Fuze 8 R.V. Axis 1	Fuze 9
T.V. -65 and M.S. -65	Fuze 10 R.V. Axis 2	Fuze 11 R.V. Axis 3	Fuze 12

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

T.V. +160 = Transportation Vibration at +160°F ± 4F°. - The units shall be tested as specified in MIL-STD-810B, Test Method 514. The following exceptions and details shall apply. The units shall be subjected to vibration over the frequency range of 5 to 500 HZ at a double amplitude of 0.5 inch or 3.5g's peak, whichever is less. Sweep time from 5 to 500 to 5 HZ shall be 15 minutes and shall be a logrithmic sweep. The total duration of vibration shall be four (4) hours in each of the three mutially perpendicular axes.

T.V. - 65 = Transportation Vibration in accordance with the T.V. +160 schedule except at -65°F ± 4F°.

M.S. +160 = Mechanical Shock shall be performed at +160°F ± 4F° after Transportation Vibration and shall consist of subjecting each fuze tested to 18 shocks, three in each of the two directions of the three mutually perpendicular planes. The shock pulse shall be 70 ± 10g peak with a 10 millisecond time duration as measured between the 10 percent points of the approximate half sine wave form.

M.S. - 65 = Same as M.S. +160 except at -65°F ± 4F°.

Following each individual environmental exposure, the S&A Device shall be removed from each fuze and the fuze, less S&A, subjected to room temperature time out test in accordance with Paragraph 4.4.4.4 with the exception that the first test shall be conducted at the fuze exposure setting and the remaining time out tests may be conducted in any sequence. Following the time out test, each unit will be set in accordance with the time out matrix prior to exposure to the next environment.

Following the last environmental exposure prior to the functional test, each unit will be subjected to the following tests at room temperature.

1. Each fuze assembly will be tested in accordance with Paragraph 3.6.6.1 and 3.6.6.2.

2. The S&A Device shall be removed and each fuze, less S&A, shall be tested for time out in accordance with Paragraph 4.4.4.4 with the exception that the first test shall be conducted at the set exposure setting prior to changing any setting and the remaining time outs may be conducted in any sequence.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the currentversion before use.

3. All S&A Devices will be tested for pin to pin resistance. In addition, Units 4 and 5 will be subjected to the time to arm test at 14g's acceleration.

4. All cover assemblies will be tested in accordance with Paragraph 3.8.15.1, 3.8.15.2 and 3.8.15.3.

5. All housing assemblies will then be tested in accordance with Paragraph 3.6.5.2, 3.6.5.3 and 3.6.5.4.

6. The individual cover modules will then be reassembled to the same original housing module.

7. All fuze assemblies, minus the S&A Device, will then be subjected to the time out test in accordance with Paragraph 4.4.4.4.

8. The S&A Devices will then be reinserted into the original housing module.

9. All fuze assemblies will then be tested in accordance with 3.6.6.1 and 3.6.6.2.

Ambient = 70°F ± 10F°

Temperature Humidity Shock = MIL-STD-331 (Test 105) Temperature Humidity with the exception that the test shall be 14 days and the transfer rate between chambers (i.e., -65°F to +160°F chamber and reverse) shall be 20 seconds maximum. Chamber temperature tolerance shall be ± 4F°.

R.F. = R.F. Testing should be accomplished at the following frequencies.

Condition 1 = 72.4 MHZ at 10 Volts/Meter with pulse modulation at 2560 HZ.

Condition 2 = 226.7 MHZ at 10 Volts/Meter with pulse modulation at 2560 HZ.

Condition 3 = 5620 MHZ at 5 Volts/Meter with pulse modulation at 2560 HZ.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

Units 3 and 6 shall be tested under Condition 1, Units 9 under Condition 2 and Unit 12 under Condition 3. Fuzes shall be functionally operated while being subjected to the R.F. signal with pulse modulation.

The fuze shall be mated to a Warhead Fuze Ring (dwg. 9283969) with the Fuze Gasket (dwg. 9267741) in place, together with Cable Assembly, XM251 (dwg. 9208657) and Gasket Fuze Ring Adapter (dwg. 9208676).

TIME OUT MATRIX

UNIT	SCO	ARM	EVENT
1	A	*10	*066.6
2	B	20	**077.7
3	C	*10	***088.8
4	D	40	199.9
5	E	***10	111.1
6	F	80	122.2
7	G	80	133.3
8	H	80	144.4
9	J	80	155.5
10	OFF	80	*100.0
11	OFF	80	**100.0
12	OFF	80	***100.0

*, **, *** - There are three (3) "zero" settings and three (3) "one" settings on the One Hundreds Event Dial. They shall be checked as indicated. There are three (3) "ten" settings on the ARM Dial.

NOTE: All units exposed to Transportation Vibration or Temperature Humidity Shock are to be considered destructively tested.

R.V. - 40
R.V. +140 = Boost Phase Random Vibration at temperature in accordance with Figure 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packing

5.1.1 Level C. - The fuze shall be packed and marked in accordance with dwg: 9220693.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.2 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100 which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.2.1.c.). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-F, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

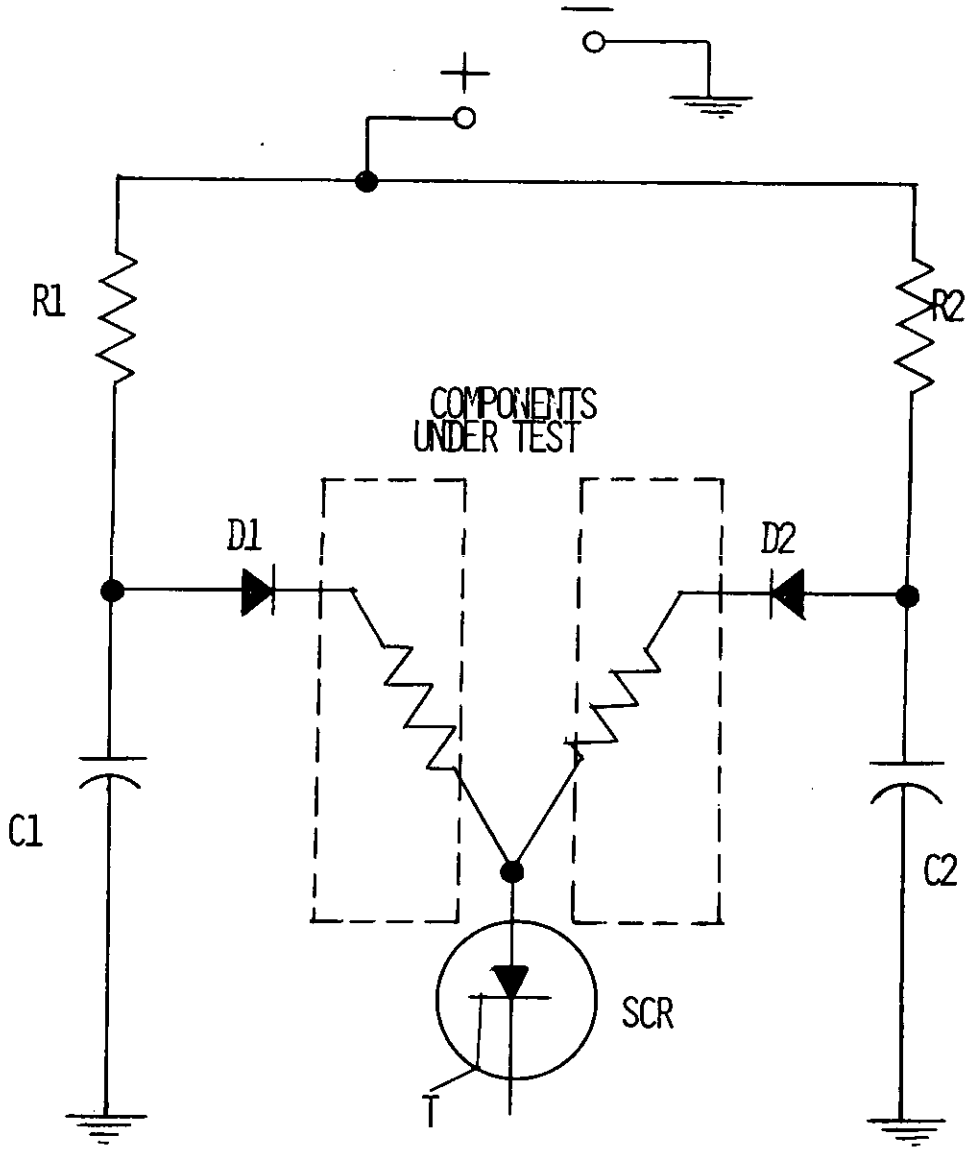
6.3 Submission of First Article Samples. - Instructions as to the location for the evaluation of the first article shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commander, Picatinny ARsenal, ATTN: SARPA-QA-I, Dover, New Jersey 07801.

Custodian:
Army-AR

Preparing Activity:
Army-PA

Project Number: 1336-A229

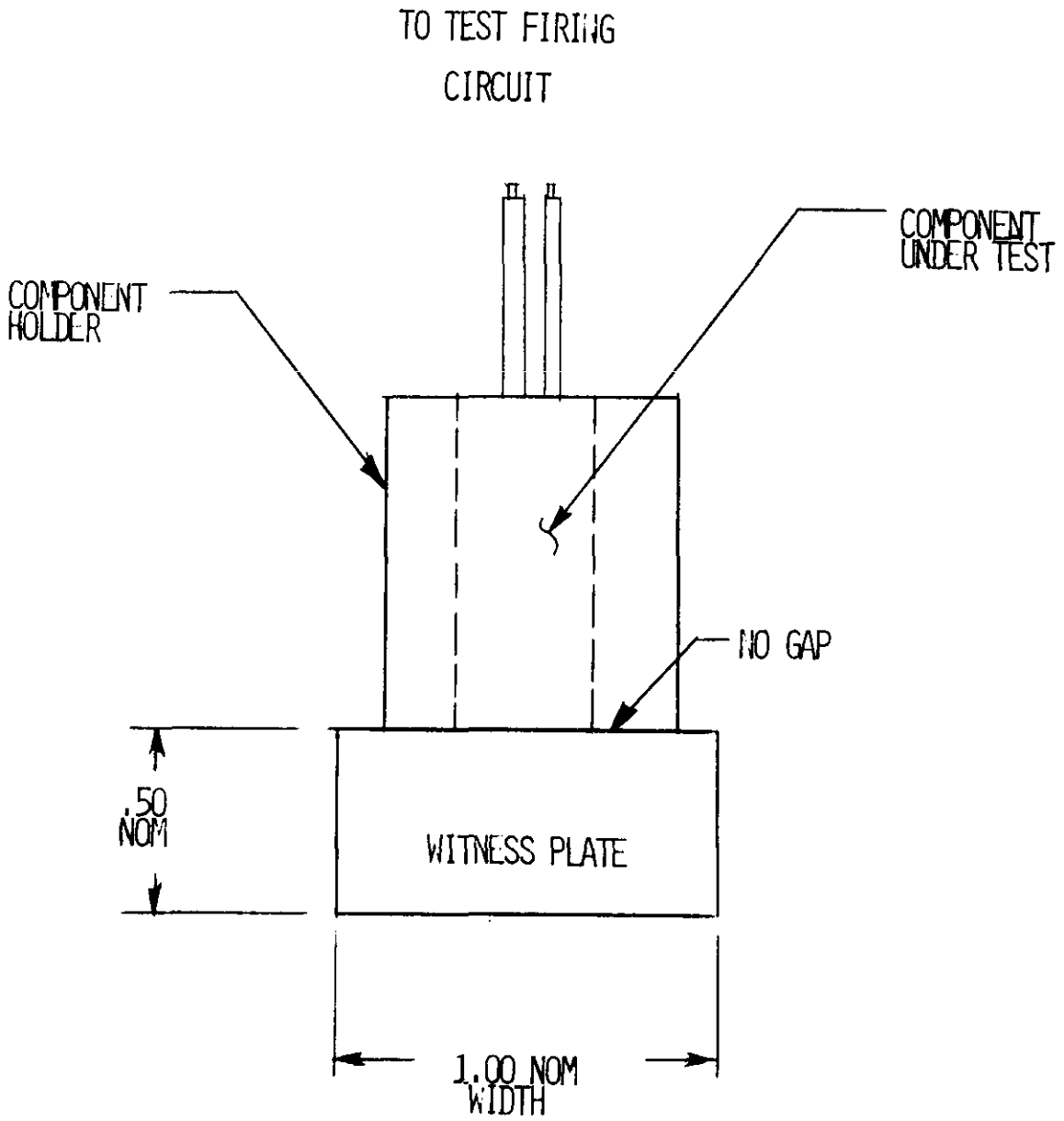
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.



DRAWING SYMBOL	P/N OR VALUE
V	TEST VOLTAGE
SCR	2N2322 (SSPI)
C1 C2	M39003/1-2266 (220 MFD ± 20% 10 VDC)
R1 R2	RCR07G103KM (10K Ω + 10/ 1/4W)
D1 D2	UT 4005 (UNITRODE)

FIGURE 2

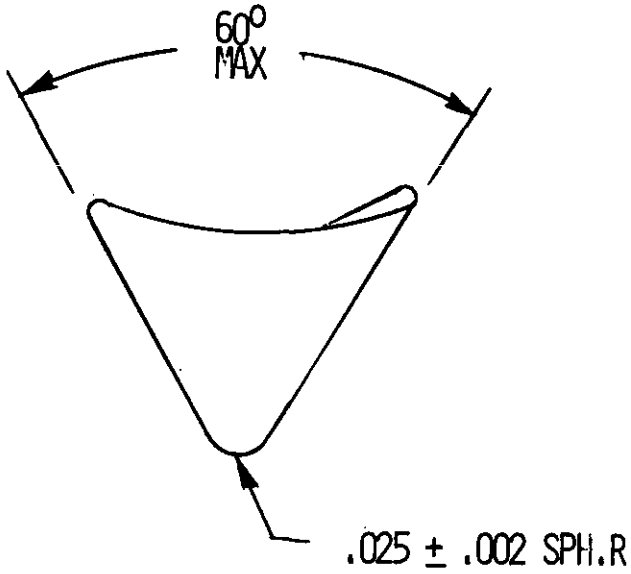
Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.



DENT TEST FIXTURE

FIG 3

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:03Z
Check the source to verify that this is the current version before use.



PROBE TIP DETAIL
FIG 4

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

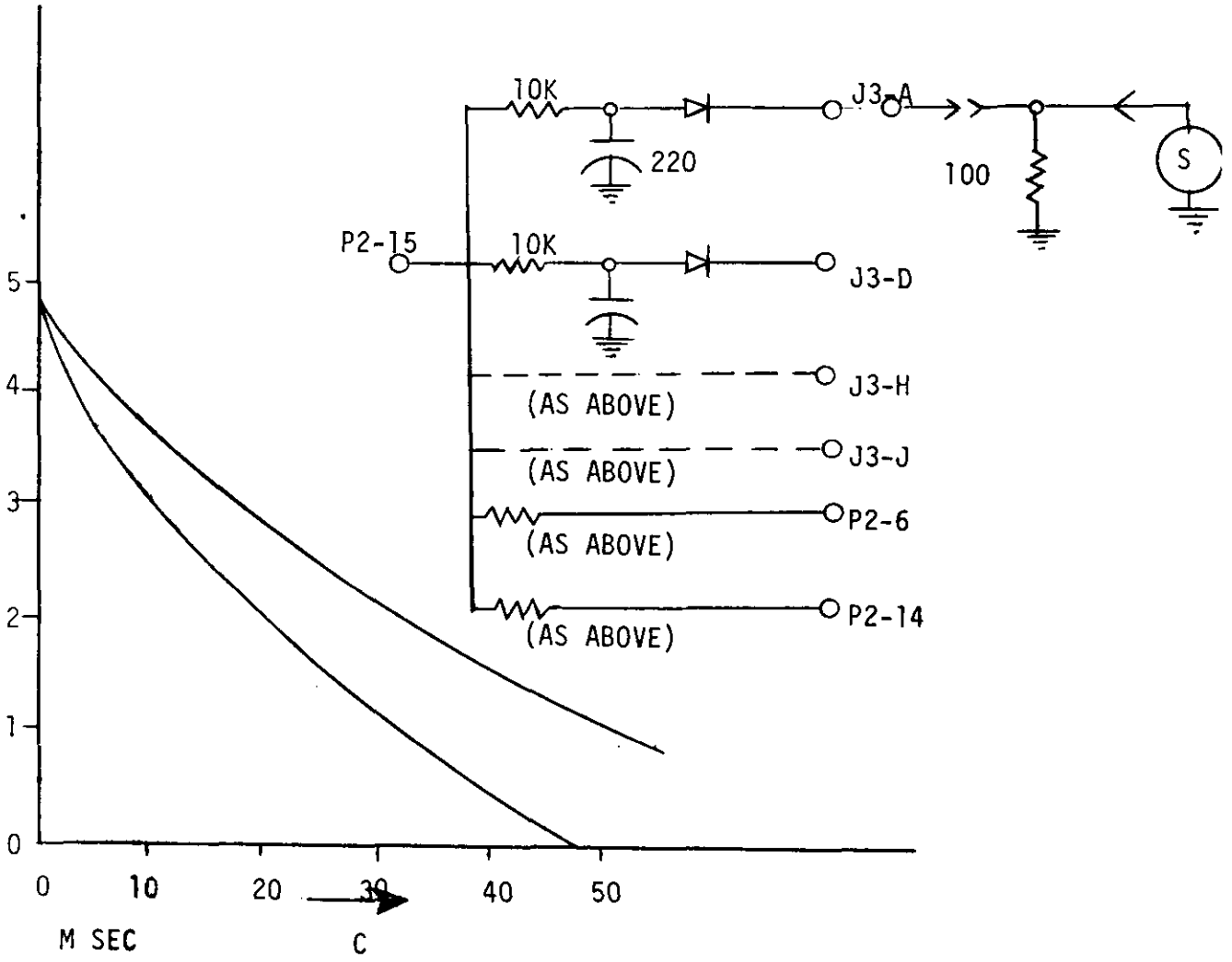


FIGURE 5 DISCHARGE PATTERN FOR ARM & EVENT CHARGING

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
Check the source to verify that this is the current version before use.

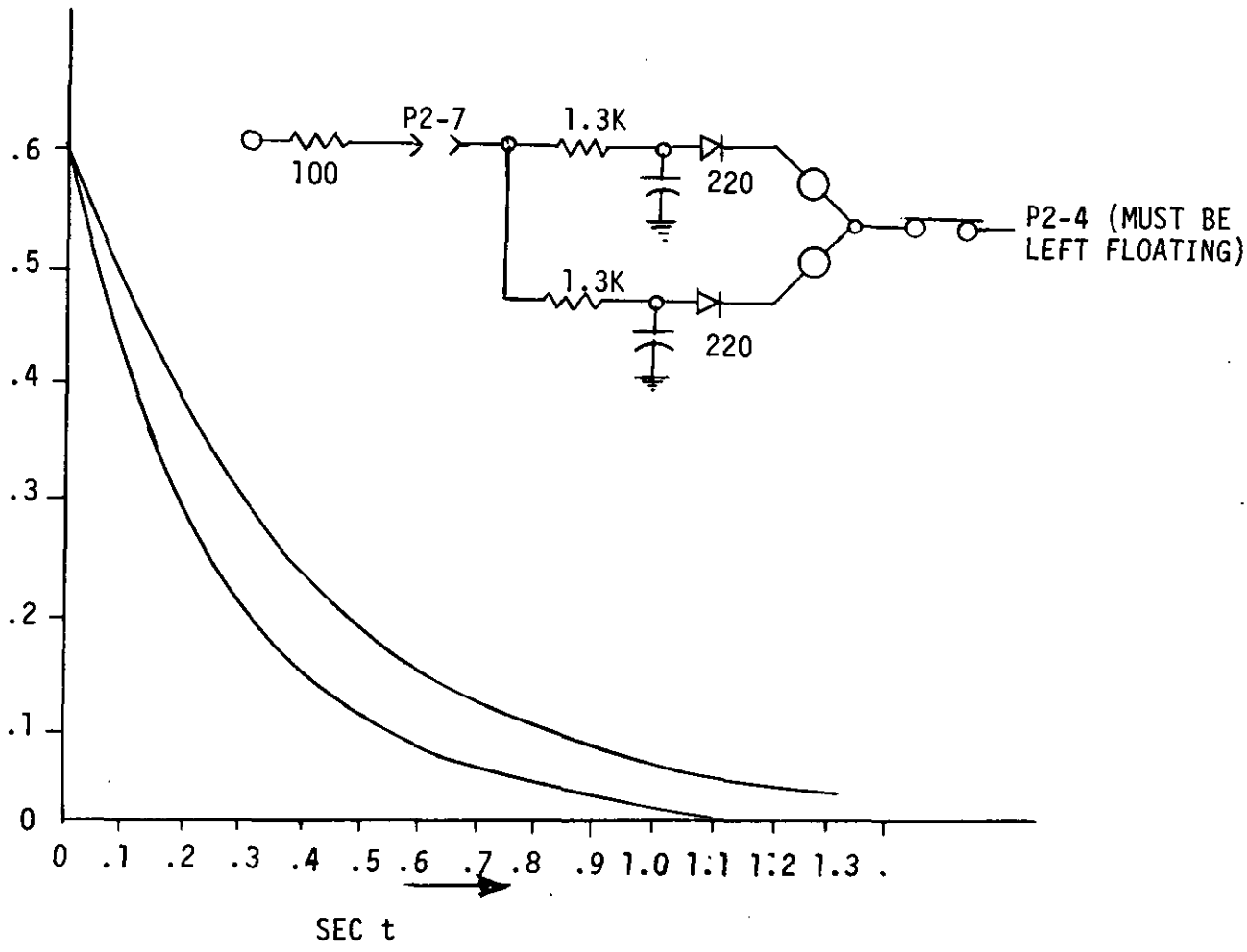


FIGURE 6 CHARGE PATTERN FOR BECO CHARGING CURRENT

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

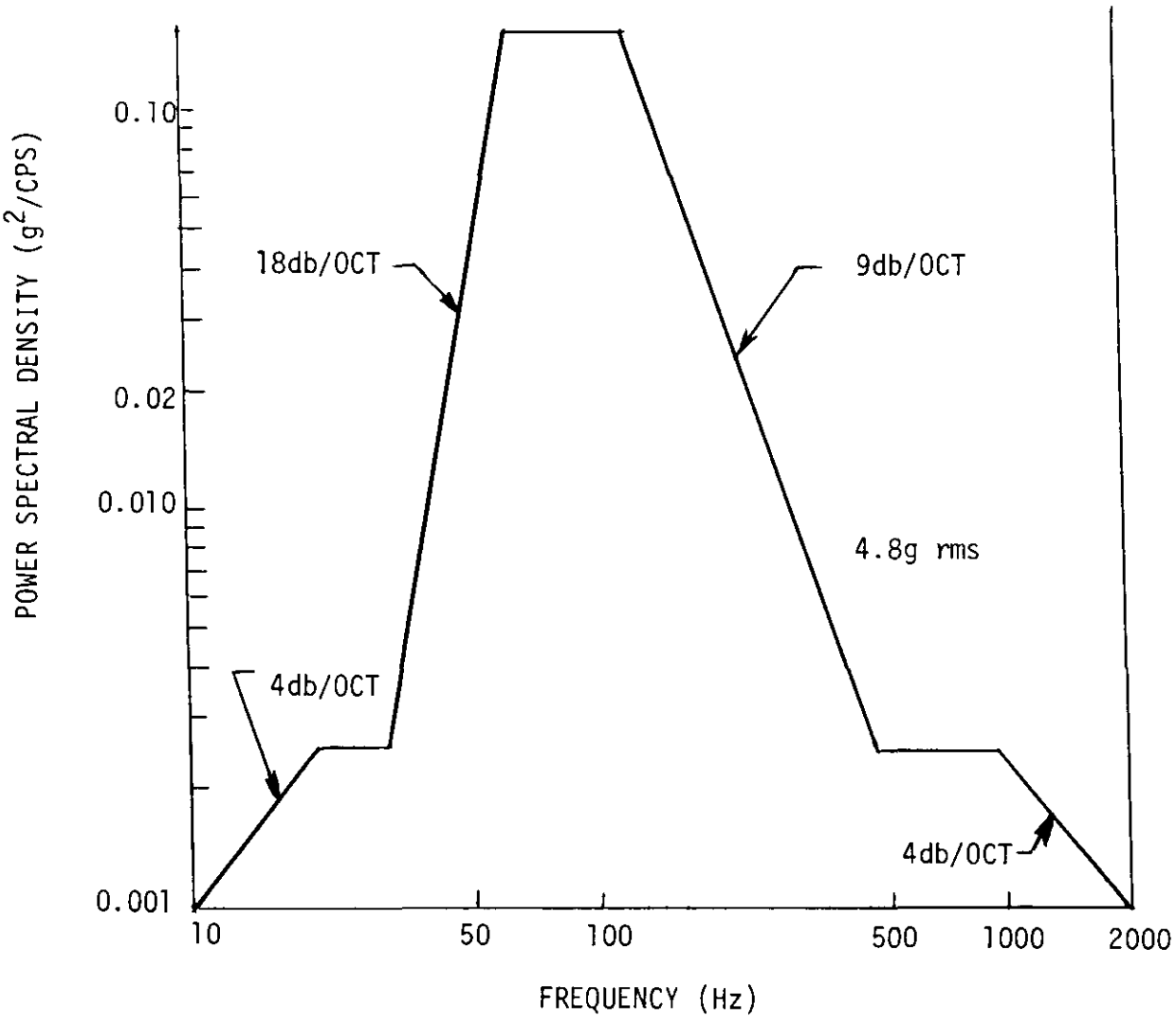


FIGURE 7, RANDOM VIBRATION SPECTRUM

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:03Z
Check the source to verify that this is the current version before use.